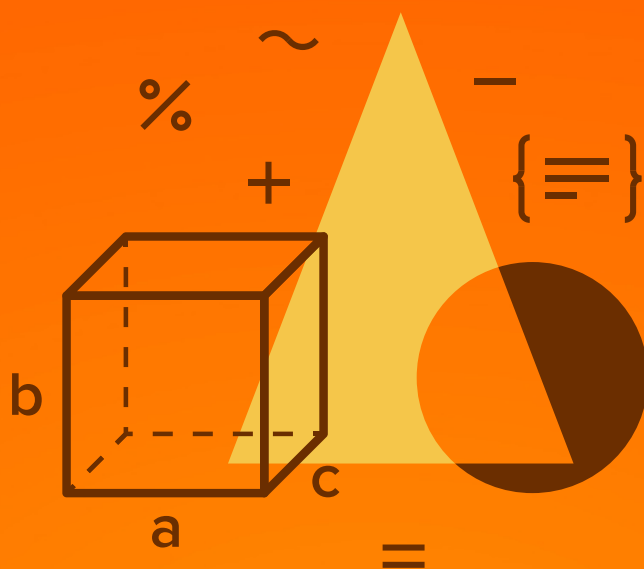




MATHEMATICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Limit, Continuity & Differentiability

English Medium

EXERCISE (O-1)

Straight Objective Type

1. If $\lim_{x \rightarrow a} \frac{2x - \sqrt{x^2 + 3a^2}}{\sqrt{x+a} - \sqrt{2a}} = \sqrt{2}$ (where $a \in \mathbb{R}^+$), then a is equal to -

- (A) $\frac{1}{3}$ (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{1}{3\sqrt{2}}$ (D) $\frac{1}{9}$

LT0005

2. $\lim_{x \rightarrow 0} \frac{\ln(\sin 3x)}{\ln(\sin x)}$ is equal to

- (A) 0 (B) 1 (C) 2 (D) Non existent

LT0006

3. $\lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x^2} - \sqrt[4]{1-2x}}{x+x^2}$ is equal to

- (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 1 (D) D.N.E.

LT0007

4. $\lim_{x \rightarrow 1} \frac{\sqrt[3]{7+x^3} - \sqrt{3+x^2}}{x-1}$ is equal to

- (A) $\frac{1}{4}$ (B) $\frac{1}{6}$ (C) $-\frac{1}{4}$ (D) $-\frac{1}{6}$

LT0008

5. $\lim_{x \rightarrow \infty} \frac{(x+1)^{10} + (x+2)^{10} + \dots + (x+100)^{10}}{x^{10} + 10^{10}}$ is equal to

- (A) 1 (B) 100 (C) 200 (D) 10

LT0010

6. $\lim_{x \rightarrow -\infty} \left(\sqrt{x^2 - 2x - 1} - \sqrt{x^2 - 7x + 3} \right)$ is equal to

- (A) $-\frac{5}{2}$ (B) $\frac{5}{2}$ (C) 0 (D) D.N.E

LT0011

7. If $\lim_{n \rightarrow \infty} \left(\sqrt{2n^2 + n} - \lambda \sqrt{2n^2 - n} \right) = \frac{1}{\sqrt{2}}$ (where λ is a real number), then-

- (A) $\lambda = 1$ (B) $\lambda = -1$ (C) $\lambda = \pm 1$ (D) $\lambda \in (-\infty, 1)$

LT0012

8. Let $U_n = \frac{n!}{(n+2)!}$ where $n \in \mathbb{N}$. If $S_n = \sum_{r=1}^n U_r$ then $\lim_{n \rightarrow \infty} S_n$ equals
 (A) 2 (B) 1 (C) 1/2 (D) Non existent **LT0013**
9. For $n \in \mathbb{N}$, let $a_n = \sum_{k=1}^n 2k$ and $b_n = \sum_{k=1}^n (2k-1)$. Then $\lim_{n \rightarrow \infty} (\sqrt{a_n} - \sqrt{b_n})$ is equal to-
 (A) 1 (B) $\frac{1}{2}$ (C) 0 (D) 2 **LT0014**
10. Let $P_n = \prod_{k=2}^n \left(1 - \frac{1}{k+1} C_2\right)$. If $\lim_{n \rightarrow \infty} P_n$ can be expressed as lowest rational in the form $\frac{a}{b}$, then value of $(a+b)$ is
 (A) 4 (B) 8 (C) 10 (D) 12 **LT0015**
11. $\lim_{x \rightarrow 1} \frac{\cos 2 - \cos 2x}{x^2 - |x|}$ is equal to
 (A) 0 (B) $\cos 2$ (C) $2\sin 2$ (D) $\sin 1$ **LT0016**
12. $\lim_{x \rightarrow 0} \left(\left[\frac{-5\sin x}{x} \right] + \left[\frac{6\sin x}{x} \right] \right)$ (where $[.]$ denotes greatest integer function) is equal to -
 (A) 0 (B) -12 (C) 1 (D) 2 **LT0017**
13. Let $f(x) = \left[\frac{\sin x}{x} \right] + \left[\frac{2\sin 2x}{x} \right] + \dots + \left[\frac{10\sin 10x}{x} \right]$ (where $[y]$ is the largest integer $\leq y$). The value of $\lim_{x \rightarrow 0} f(x)$ equals
 (A) 55 (B) 164 (C) 165 (D) 375 **LT0018**
14. Let $f(x) = \frac{\sin\{x\}}{x^2 + ax + b}$. If $f(5^+)$ & $f(3^+)$ exists finitely and are not zero, then the value of $(a+b)$ is
 (where $\{.\}$ represents fractional part function) -
 (A) 7 (B) 10 (C) 11 (D) 20 **LT0019**
15. $\lim_{x \rightarrow 0} \frac{|\cos(\sin(3x))| - 1}{x^2}$ equals
 (A) $-\frac{9}{2}$ (B) $-\frac{3}{2}$ (C) $\frac{3}{2}$ (D) $\frac{9}{2}$ **LT0020**

16. Let $a = \min\{x^2 + 2x + 3, x \in \mathbb{R}\}$ and $b = \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta^2}$. Then value of $\sum_{r=0}^n a^r \cdot b^{n-r}$ is :

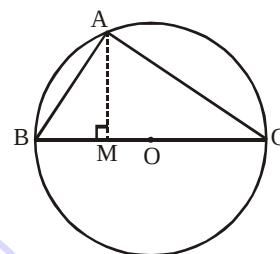
- (A) $\frac{2^{n+1} - 1}{3 \cdot 2^n}$ (B) $\frac{2^{n+1} + 1}{3 \cdot 2^n}$ (C) $\frac{4^{n+1} - 1}{3 \cdot 2^n}$ (D) N.O.T.

LT0021

17. Let BC is diameter of a circle centred at O. Point A is a variable point, moving on the circumference of circle. If BC = 1 unit, then

$\lim_{A \rightarrow B} \frac{BM}{(\text{Area of sector OAB})^2}$ is equal to -

- (A) 1 (B) 2
(C) 4 (D) 16



LT0022

18. $\lim_{x \rightarrow \infty} \left(\frac{x^2 - 2x + 1}{x^2 - 4x + 2} \right)^x$ is equal to

- (A) 1 (B) e (C) $\frac{1}{e^2}$ (D) e^2

LT0023

19. $\lim_{x \rightarrow 0} (1 + \sin x)^{\cos x}$ is equal to

- (A) 0 (B) e (C) 1 (D) $\frac{1}{e}$

LT0024

20. $\lim_{x \rightarrow 0} (\cos x + a \sin bx)^{\frac{1}{x}}$ is equal to

- (A) e^a (B) e^{ab} (C) e^b (D) $e^{a/b}$

LT0025

21. $\lim_{x \rightarrow 0} \left(\tan \left(\frac{\pi}{4} + x \right) \right)^{1/x}$ is equal to

- (A) e^{-2} (B) $\frac{1}{e}$ (C) e (D) e^2

LT0026

22. $\lim_{n \rightarrow \infty} (4^n + 5^n)^{1/n}$ is equal to

- (A) 5 (B) 4 (C) 0 (D) D.N.E.

LT0027

23. $\lim_{x \rightarrow \infty} \left(\frac{1^{1/x} + 2^{1/x} + 3^{1/x} + \dots + n^{1/x}}{n} \right)^{nx}$, $n \in \mathbb{N}$ is equal to

- (A) $n!$ (B) 1 (C) $\frac{1}{n!}$ (D) 0

LT0028

24. If $\lim_{x \rightarrow \lambda} \left(2 - \frac{\lambda}{x} \right)^{\lambda \tan\left(\frac{\pi x}{2\lambda}\right)} = \frac{1}{e}$, then λ is equal to -

- (A) $-\pi$ (B) π (C) $\frac{\pi}{2}$ (D) $-\frac{2}{\pi}$

LT0029

25. If $\lim_{x \rightarrow 0} (1 + ax + bx^2)^{2/x} = e^3$, then

- (A) $a = \frac{3}{2}$ and $b \in \mathbb{R}$ (B) $a = \frac{3}{2}$ and $b \in \mathbb{R}^+$
(C) $a = 0$ and $b = 1$ (D) $a = 1$ and $b = 0$

LT0030

26. If $f(x)$ is a polynomial of least degree, such that $\lim_{x \rightarrow 0} \left(1 + \frac{f(x) + x^2}{x^2} \right)^{1/x} = e^2$, then $f(2)$ is -

- (A) 2 (B) 8 (C) 10 (D) 12

LT0031

27. Let $f(x) = \frac{\tan x}{x}$, then the value of $\lim_{x \rightarrow 0} \left([f(x)] + x^2 \right)^{\frac{1}{\{f(x)\}}}$ is equal to (where $[\cdot]$, $\{\cdot\}$ denotes greatest integer function and fractional part functions respectively) -

(A) e^{-3} (B) e^3 (C) e^2 (D) non-existent

LT0032

28. $\lim_{n \rightarrow \infty} \frac{e^n}{\left(1 + \frac{1}{n} \right)^{n^2}}$ equals -

- (A) 1 (B) $\frac{1}{2}$ (C) e (D) $\sqrt{e}$

LT0033

29. If $f(x)$ is odd linear polynomial with $f(1) = 1$, then $\lim_{x \rightarrow 0} \frac{2^{f(\tan x)} - 2^{f(\sin x)}}{x^2 f(\sin x)}$ is

- (A) 1 (B) $\ln 2$ (C) $\frac{1}{2} \ln 2$ (D) $\cos 2$

LT0034

30. $\lim_{x \rightarrow 0} \frac{x(1 + a \cos x) - b \sin x}{x^3} = 1$ then

- (A) $a = -5/2$ (B) $a = -3/2$, $b = -1/2$
(C) $a = -3/2$, $b = -5/2$ (D) $a = -5/2$, $b = -3/2$

LT0035

EXERCISE (O-2)

Multiple Correct Answer Type

1. Consider following statements and identify correct options

$$(i) \lim_{x \rightarrow 4} \left(\frac{2x}{x-4} - \frac{8}{x-4} \right) = \lim_{x \rightarrow 4} \frac{2x}{x-4} - \lim_{x \rightarrow 4} \frac{8}{x-4}$$

$$(ii) \lim_{x \rightarrow 1} \frac{x^2 + 6x - 7}{x^2 + 5x - 6} = \frac{\lim_{x \rightarrow 1} (x^2 + 6x - 7)}{\lim_{x \rightarrow 1} (x^2 + 5x - 6)}$$

$$(iii) \lim_{x \rightarrow 1} \frac{x-3}{x^2 + 2x - 4} = \frac{\lim_{x \rightarrow 1} (x-3)}{\lim_{x \rightarrow 1} (x^2 + 2x - 4)}$$

(iv) If $\lim_{x \rightarrow 5} f(x) = 2$ and $\lim_{x \rightarrow 5} g(x) = 0$, then $\lim_{x \rightarrow 5} \frac{f(x)}{g(x)}$ does not exist.

(v) If $\lim_{x \rightarrow 5} f(x) = 0$ and $\lim_{x \rightarrow 5} g(x) = 2$, then $\lim_{x \rightarrow 5} \frac{f(x)}{g(x)}$ does not exist.

(A) Only one is true.

(B) Only two are true.

(C) Only three are false.

(D) Only two are false.

LT0036

2. Which of the following limits equal to $\frac{1}{2}$

$$(A) \lim_{n \rightarrow \infty} \left(\frac{1}{1.3} + \frac{1}{3.5} + \dots + \frac{1}{(2n-1)(2n+1)} \right)$$

$$(B) \lim_{x \rightarrow \infty} \left[\frac{3x^2}{2x+1} - \frac{(2x-1)(3x^2+x+2)}{4x^2} \right]$$

$$(C) \lim_{n \rightarrow \infty} \frac{1}{n^2} (1 + 2 + 3 + \dots + n)$$

$$(D) \lim_{n \rightarrow \infty} \frac{(n+2)! + (n+1)!}{(n+2)! - (n+1)!}$$

LT0037

3. Let $f(x) = \begin{cases} \sin x & \text{where } x = \text{integer} \\ 0 & \text{otherwise} \end{cases}$: $g(x) = \begin{cases} x^2 + 1 & ; x \neq 0, 2 \\ 4 & ; x = 0 \\ 5 & ; x = 2 \end{cases}$, then

$$(A) \lim_{x \rightarrow 0} g(f(x)) = 4$$

$$(B) \lim_{x \rightarrow 0} f(g(x)) = 0$$

$$(C) \lim_{x \rightarrow 1} f(g(x)) = 0$$

$$(D) \lim_{x \rightarrow 1} g(f(x)) = 5$$

LT0038

4. If $f(x) = \begin{cases} x^2 & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$, then

$$(A) \lim_{x \rightarrow 0} f(x) = 0$$

$$(B) \lim_{x \rightarrow 0} f(x) \text{ does not exist}$$

$$(C) \lim_{x \rightarrow 2} f(x) = 4$$

$$(D) \lim_{x \rightarrow 2} f(x) \text{ does not exist}$$

LT0039

5. Let $f(\beta) = \lim_{\alpha \rightarrow \beta} \frac{\sin^2 \alpha - \sin^2 \beta}{\alpha^2 - \beta^2}$, then $f\left(\frac{\pi}{4}\right)$ is greater than-

(A) $\lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{x \sin 2x}$

(B) $\lim_{x \rightarrow \pi/2} \frac{\cot x - \cos x}{(\pi - 2x)^3}$

(C) $\lim_{x \rightarrow \infty} (\cos \sqrt{x+1} - \cos \sqrt{x})$

(D) $\lim_{x \rightarrow a} \frac{\sqrt{a+2x} - \sqrt{3x}}{\sqrt{3a+x} - 2\sqrt{x}}$ where $a > 0$

LT0040

6. If $\frac{\sin x + ae^x + be^{-x} + \ln(1+x)}{x^3}$ has a finite limit L as $x \rightarrow 0$, then

(A) $a = -\frac{1}{2}$

(B) $b = \frac{1}{2}$

(C) $c = 0$

(D) $L = -\frac{1}{3}$

LT0041

7. Let $\ell = \lim_{x \rightarrow \infty} \frac{a^x - a^{-x}}{a^x + a^{-x}}$ ($a > 0$), then

(A) $\ell = 1 \forall a > 0$

(B) $\ell = -1 \forall a \in (0, 1)$

(C) $\ell = 0$, if $a = 1$

(D) $\ell = 1 \forall a > 1$

LT0042

8. If $\ell = \lim_{x \rightarrow a} \frac{\sqrt{3x^2 + a^2} - \sqrt{x^2 + 3a^2}}{(x-a)}$ then -

(A) $\ell = 1 \forall a \in \mathbb{R}$

(B) $\ell = 1 \forall a > 0$

(C) $\ell = -1 \forall a < 0$

(D) $\ell = \text{D.N.E.}$ if $a = 0$

LT0058

9. Which of the following limits vanish?

(A) $\lim_{x \rightarrow \infty} \frac{\sin x}{x}$

(B) $\lim_{x \rightarrow \infty} \frac{\arctan x}{x}$

(C) $\lim_{x \rightarrow \infty} \frac{x + \sin x}{x + \cos x}$

(D) $\lim_{x \rightarrow 1} \frac{\arcsin x}{\tan \frac{\pi x}{2}}$

LT0059

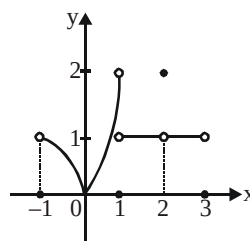
10. Which of the following statement are true for the function f defined for $-1 \leq x \leq 3$ in the figure shown.

(A) $\lim_{x \rightarrow -1^+} f(x) = 1$

(B) $\lim_{x \rightarrow 2} f(x)$ does not exist

(C) $\lim_{x \rightarrow 1^-} f(x) = 2$

(D) $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x)$



LT0060

11. Let $f(x) = x + \sqrt{x^2 + 2x}$ and $g(x) = \sqrt{x^2 + 2x} - x$, then

(A) $\lim_{x \rightarrow \infty} g(x) = 1$

(B) $\lim_{x \rightarrow \infty} f(x) = 1$

(C) $\lim_{x \rightarrow -\infty} f(x) = -1$

(D) $\lim_{x \rightarrow -\infty} g(x) = -1$

LT0061

12. If $A = \lim_{x \rightarrow 0} \frac{\sin^{-1}(\sin x)}{\cos^{-1}(\cos x)}$ and $B = \lim_{x \rightarrow 0} \frac{[x]}{x}$, then (where $[.]$ denotes greatest integer function)-

(A) $A = 1$

(B) A does not exist

(C) $B = 0$

(D) $B = 1$

LT0062

13. Which of the following limit tends to unity ?

(A) $\lim_{x \rightarrow 0} \frac{1 - \cos x + 2 \sin x - \sin^3 x - x^2 + 3x^4}{\tan^3 x - 6 \sin^2 x + x - 5x^3}$

(B) $\lim_{x \rightarrow \infty} \frac{x}{[x]}$

(C) $\lim_{x \rightarrow \infty} \frac{1}{(\sqrt{x} + \sqrt{x + \sqrt{x}} - \sqrt{x})}$

(D) $\lim_{x \rightarrow \infty} \left(\frac{\sqrt{x}}{\sqrt{x} + \sqrt{x + \sqrt{x}}} \right)$

LT0063

14. Which of the following limits does not exist ?

(A) $\lim_{x \rightarrow 1^+} ([x])^{\frac{1}{x-1}}$

(B) $\lim_{x \rightarrow 3} \frac{(x^2 - 9 - \sqrt{x^2 - 6x + 9})}{|x - 1| - 2}$

(C) $\lim_{x \rightarrow 0^+} (x)^{\ln x}$

(D) $\lim_{x \rightarrow 0^+} \left(\frac{1 - \cos(\sin^2 x)}{x^2} \right)^{\frac{\ln(1-2x^2)}{\sin^2 x}}$

(where $[.]$ represents greatest integer function)

LT0064

15. The value(s) of 'n' for which $\lim_{x \rightarrow 1} \frac{e^{x-1} - x}{(x-1)^n}$ exists is/are -

(A) 1

(B) 2

(C) 3

(D) 4

LT0065

16. Let $f: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$, $f(x) = \begin{cases} \lim_{n \rightarrow \infty} \left(\frac{(\tan x)^{2n} + x^2}{\sin^2 x + (\tan x)^{2n}} \right); & x \neq 0 \\ 1 & ; x = 0 \end{cases}$, $n \in \mathbb{N}$. Which of the following holds

good ?

(A) $f\left(-\frac{\pi}{4}\right) = f\left(\frac{\pi}{4}\right)$

(B) $f\left(-\frac{\pi}{4}\right) = f\left(-\frac{\pi}{4}\right)$

(C) $f\left(\frac{\pi}{4}\right) = f\left(\frac{\pi}{4}\right)$

(D) $f(0^+) = f(0) = f(0^-)$

LT0066

17. Let $f(x) = \begin{cases} \frac{\tan^2 \{x\}}{x^2 - [x]^2} & \text{for } x > 0 \\ 1 & \text{for } x = 0 \\ \sqrt{\{x\} \cot \{x\}} & \text{for } x < 0 \end{cases}$, where $[x]$ is the step up function and $\{x\}$ is the fractional part function of x , then-

(A) $\lim_{x \rightarrow 0^+} f(x) = 1$

(B) $\lim_{x \rightarrow 0^-} f(x) = 1$

(C) $\cot^{-1} \left(\lim_{x \rightarrow 0^-} f(x) \right)^2 = 1$

(D) None

LT0067

18. $\lim_{x \rightarrow c} f(x)$ does not exist when (where $[x]$ is the step up function, $\{x\}$ is the fractional part function of x & $\text{sgn}(x)$ denotes signum function), then-

(A) $f(x) = [[x]] - [2x - 1]$; $c = 3$

(B) $f(x) = [x] - x$, $c = 1$

(C) $f(x) = \{x\}^2 - \{-x\}^2$, $c = 0$

(D) $f(x) = \frac{\tan(\text{sgn } x)}{\text{sgn } x}$, $c = 0$

LT0068

19. Which of the following limits does not exist ?

(A) $\lim_{x \rightarrow \infty} \text{cosec}^{-1} \left(\frac{x}{x+7} \right)$

(B) $\lim_{x \rightarrow 1} \text{sec}^{-1}(\sin^{-1} x)$

(C) $\lim_{x \rightarrow 0^+} x^{\frac{1}{x}}$

(D) $\lim_{x \rightarrow 0} \left(\tan \left(\frac{\pi}{8} + x \right) \right)^{\cot x}$

LT0069

20. Which of the following statement(s) is (are) INCORRECT ?

(A) If $\lim_{x \rightarrow c} f(x)$ and $\lim_{x \rightarrow c} g(x)$ both does not exist then $\lim_{x \rightarrow c} f(x)g(x)$ also does not exist.

(B) If $\lim_{x \rightarrow c} f(x)$ and $\lim_{x \rightarrow c} g(x)$ both does not exist then $\lim_{x \rightarrow c} f(g(x))$ also does not exist.

(C) If $\lim_{x \rightarrow c} f(x)$ exists and $\lim_{x \rightarrow c} g(x)$ does not exist then $\lim_{x \rightarrow c} g(f(x))$ does not exist.

(D) If $\lim_{x \rightarrow c} f(x)$ and $\lim_{x \rightarrow c} g(x)$ both exist then $\lim_{x \rightarrow c} f(g(x))$ and $\lim_{x \rightarrow c} g(f(x))$ also exist.

LT0070

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 to 3

Consider two functions $f(x) = \lim_{n \rightarrow \infty} \left(\cos \frac{x}{\sqrt{n}} \right)^n$ and $g(x) = -x^{4b}$,

where $b = \lim_{x \rightarrow \infty} \left(\sqrt{x^2 + x + 1} - \sqrt{x^2 + 1} \right)$, then

1. $f(x)$ is LT0143
 (A) e^{-x^2} (B) $e^{\frac{-x^2}{2}}$ (C) e^{x^2} (D) $e^{\frac{x^2}{2}}$
2. $g(x)$ is LT0144
 (A) $-x^2$ (B) x^2 (C) x^4 (D) $-x^4$
3. Number of solutions of equation $f(x) + g(x) = 0$ is LT0145
 (A) 2 (B) 4 (C) 0 (D) 1

Paragraph for Question 4 to 6

Consider $f(x) = \frac{\sin x + ae^x + be^{-x} + c \ln(1+x)}{x^3}$, where a, b, c are real numbers.

4. If $\lim_{x \rightarrow 0^+} f(x)$ is finite, then the value of $a - b + c$ is LT0146
5. If $\lim_{x \rightarrow 0^+} f(x) = \ell$ (finite), then the value of ℓ is LT0147
6. Using the values of a, b, c as found above, the value of $\lim_{x \rightarrow 0^+} x f(x)$ is LT0148

Matrix Match Type

7. $\lim_{x \rightarrow 0} f(x)$, where $f(x)$ is as in LIST-I, is: LT0149

List-I

(P) $f(x) = \frac{\tan[e^2]x^2 - \tan[-e^2]x^2}{\sin^2 x}$

where $[.]$ represents greatest integer function.

(Q) $f(x) = \left[\left(\min(t^2 + 4t + 6) \right) \frac{\sin x}{x} \right]$

where $[.]$ represents greatest integer function.

(R) $f(x) = \frac{\sqrt[3]{1+x^2} - \sqrt[4]{1-2x}}{x+x^2}$

(S) $f(x) = \frac{\sqrt{2} - \sqrt{1+\cos x}}{\sin^2 x}$

List-II

(1) $\frac{\sqrt{2}}{8}$

(2) 15

(3) 1

(4) $\frac{1}{2}$

- (A) (P) $\rightarrow$ (3), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (4), (S) $\rightarrow$ (1)
 (B) (P) $\rightarrow$ (2), (Q) $\rightarrow$ (3), (R) $\rightarrow$ (4), (S) $\rightarrow$ (1)
 (C) (P) $\rightarrow$ (2), (Q) $\rightarrow$ (3), (R) $\rightarrow$ (1), (S) $\rightarrow$ (4)
 (D) (P) $\rightarrow$ (2), (Q) $\rightarrow$ (4), (R) $\rightarrow$ (3), (S) $\rightarrow$ (1)

8. For the function $g(t)$ whose graph is given, match the entries of column-I to column-II.

Column-I

(A) $\lim_{t \rightarrow 0^+} g(t) + \lim_{t \rightarrow 2^-} g(t)$

(B) $\lim_{t \rightarrow 0^-} g(t) + g(2)$

(C) $\lim_{t \rightarrow 0} g(t)$

(D) $\lim_{t \rightarrow 2} g(t)$

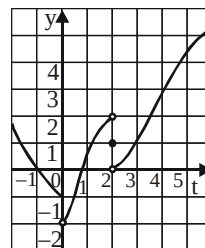
Column-II

(P) $\lim_{t \rightarrow 2^+} g(t)$

(Q) does not exist

(R) 0

(S) $\lim_{t \rightarrow 4} g(t)$



LT0043

9.

Column-I

(A) $\lim_{n \rightarrow \infty} n \sin\left(\frac{\pi}{4n}\right) \cos\left(\frac{\pi}{4n}\right)$ is equal to

(B) $\lim_{x \rightarrow 0} \frac{\sin x^\circ}{x}$ is equal to

(C) $\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{\tan x} \right)$ is equal to

(D) $\lim_{x \rightarrow \pi/2} \frac{1 + \cos 2x}{(\pi - 2x)^2}$ is equal to

Column-II

(P) 0

(Q) $\frac{1}{2}$

(R) $\frac{\pi}{4}$

(S) $\frac{\pi}{180}$

LT0044

10.

Column-I

(A) $\lim_{x \rightarrow \infty} \frac{a^x}{a^x + 1}$ ($a > 0$) can be equal to

(B) $\lim_{x \rightarrow 2} \frac{\sin(e^{x-2} - 1)}{\log(x - 1)}$ is equal to

(C) $\lim_{x \rightarrow e} \frac{(\ln x - 1)e}{x - e}$ is equal to

(D) $\lim_{x \rightarrow 0} \frac{x(5^x - 1)}{(1 - \cos x)4 \ln 5}$ is equal to

Column-II

(P) $\lim_{x \rightarrow \infty} x(e^{1/x} - 1)$

(Q) $\lim_{x \rightarrow 0} \frac{a^x + b^x + c^x - 3}{x}$ ($a, b, c > 0$ & $abc = 1$)

(R) $\lim_{x \rightarrow 0} \frac{e^{4x} - e^{3x}}{x}$

(S) $\frac{1}{2}$

(T) 0

LT0045

EXERCISE (O-4)

Numerical Grid Type

1. $\lim_{x \rightarrow 1} \frac{x^2 - x \cdot \ln x + \ln x - 1}{x - 1}$ LT0071
2. $\lim_{x \rightarrow 0} [\ell n (1 + \sin^2 x) \cdot \cot (\ell n^2 (1 + x))]$ LT0081
3. If $\lim_{x \rightarrow \infty} \frac{a(2x^3 - x^2) + b(x^3 + 5x^2 - 1) - c(3x^3 + x^2)}{a(5x^4 - x) - bx^4 + c(4x^4 + 1) + 2x^2 + 5x} = 1$, then the value of $(a + b + c)$ can be expressed in the lowest form as $\frac{p}{q}$. Find the value of $(p + q)$. LT0094
4. If $L = \lim_{\theta \rightarrow \frac{\pi}{4}} \frac{\sqrt{2} - \cos \theta - \sin \theta}{(4\theta - \pi)^2}$, then the value of $8\sqrt{2}L$ is LT0151
5. $\lim_{h \rightarrow 0} \frac{\sin\left(\frac{\pi}{3} + 4h\right) - 4\sin\left(\frac{\pi}{3} + 3h\right) + 6\sin\left(\frac{\pi}{3} + 2h\right) - 4\sin\left(\frac{\pi}{3} + h\right) + \sin \frac{\pi}{3}}{\sqrt{3} \cdot h^4}$ LT0152
6. $\lim_{x \rightarrow \infty} x^2 \left(\sqrt{\frac{x+2}{x}} - \sqrt[3]{\frac{x+3}{x}} \right)$ LT0077
7. $\lim_{x \rightarrow -\infty} \frac{(3x^4 + 2x^2) \sin \frac{1}{x} + |x|^3 + 5}{|x|^3 + |x|^2 + |x| + 1}$ LT0078
8. If $\lim_{x \rightarrow -\infty} \left[\sqrt{x^2 - x + 1} - ax - b \right] = 0$, then the value of $(a + b)$ is LT0150
9. Let $f(x) = ax^3 + bx^2 + cx + d$ and $g(x) = x^2 + x - 2$.
If $\lim_{x \rightarrow 1} \frac{f(x)}{g(x)} = 1$ and $\lim_{x \rightarrow -2} \frac{f(x)}{g(x)} = 4$, then find the value of $\frac{c^2 + d^2}{a^2 + b^2}$. LT0085
10. Let $\{a_n\}$, $\{b_n\}$, $\{c_n\}$ be sequences such that
(i) $a_n + b_n + c_n = 2n + 1$;
(ii) $a_n b_n + b_n c_n + c_n a_n = 2n - 1$;
(iii) $a_n b_n c_n = -1$;
(iv) $a_n < b_n < c_n$
Then find the value of $\lim_{n \rightarrow \infty} (na_n)$. LT0084

EXERCISE (JM)

1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a positive increasing function with $\lim_{x \rightarrow \infty} \frac{f(3x)}{f(x)} = 1$. Then $\lim_{x \rightarrow \infty} \frac{f(2x)}{f(x)} =$

[AIEEE-2010]

- (1) 1 (2) $\frac{2}{3}$ (3) $\frac{3}{2}$ (4) 3

LT0107

2. $\lim_{x \rightarrow 2} \left(\frac{\sqrt{1 - \cos\{2(x-2)\}}}{x-2} \right)$

[AIEEE-2011]

- (1) equals $-\sqrt{2}$ (2) equals $\frac{1}{\sqrt{2}}$ (3) does not exist (4) equals $\sqrt{2}$

LT0108

3. Let $f : \mathbb{R} \rightarrow [0, \infty)$ be such that $\lim_{x \rightarrow 5} f(x)$ exists and $\lim_{x \rightarrow 5} \frac{(f(x))^2 - 9}{\sqrt{|x-5|}} = 0$. Then $\lim_{x \rightarrow 5} f(x)$ equal -

[AIEEE-2011]

- (1) 3 (2) 0 (3) 1 (4) 2

LT0109

4. $\lim_{x \rightarrow 0} \frac{\sin(\pi \cos^2 x)}{x^2}$ is equal to :

[JEE Mains Offline-2014]

- (1) $\frac{\pi}{2}$ (2) 1 (3) $-\pi$ (4) π

LT0110

5. If $\lim_{x \rightarrow 2} \frac{\tan(x-2)\{x^2 + (k-2)x - 2k\}}{x^2 - 4x + 4} = 5$ then k is equal to

[JEE Mains Online-2014]

- (1) 3 (2) 1 (3) 0 (4) 2

LT0111

6. Let $p = \lim_{x \rightarrow 0+} \left(1 + \tan^2 \sqrt{x} \right)^{\frac{1}{2x}}$ then $\log p$ is equal to -

[JEE(Main)-2016]

- (1) $\frac{1}{4}$ (2) 2 (3) 1 (4) $\frac{1}{2}$

LT0112

7. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cot x - \cos x}{(\pi - 2x)^3}$ equals :-

[JEE(Main)-2017]

- (1) $\frac{1}{4}$ (2) $\frac{1}{24}$ (3) $\frac{1}{16}$ (4) $\frac{1}{8}$

LT0113

8. For each $t \in \mathbb{R}$, let $[t]$ be the greatest integer less than or equal to t . Then

$$\lim_{x \rightarrow 0^+} x \left(\left[\frac{1}{x} \right] + \left[\frac{2}{x} \right] + \dots + \left[\frac{15}{x} \right] \right)$$

[JEE(Main)-2018]

- (1) is equal to 15. (2) is equal to 120.
(3) does not exist (in $\mathbb{R}$). (4) is equal to 0.

LT0114

9.
$$\lim_{y \rightarrow 0} \frac{\sqrt{1 + \sqrt{1 + y^4}} - \sqrt{2}}{y^4}$$

[JEE(Main)-2019]

- (1) exists and equals $\frac{1}{4\sqrt{2}}$ (2) does not exist
(3) exists and equals $\frac{1}{2\sqrt{2}}$ (4) exists and equals $\frac{1}{2\sqrt{2}(\sqrt{2}+1)}$

LT0115

10. For each $x \in \mathbb{R}$, let $[x]$ be the greatest integer less than or equal to x .

Then $\lim_{x \rightarrow 0^-} \frac{x([x] + |x|) \sin[x]}{|x|}$ is equal to

[JEE(Main)-2019]

- (1) $-\sin 1$ (2) 0 (3) 1 (4) $\sin 1$

LT0116

11. For each $t \in \mathbb{R}$, let $[t]$ be the greatest integer less than or equal to t .

Then,
$$\lim_{x \rightarrow 1^+} \frac{(1 - |x| + \sin |1 - x|) \sin \left(\frac{\pi}{2} [1 - x] \right)}{|1 - x| [1 - x]}$$

[JEE(Main)-2019]

- (1) equals -1 (2) equals 1 (3) does not exist (4) equals 0

LT0117

12. Let $[x]$ denote the greatest integer less than or equal to x .

Then
$$\lim_{x \rightarrow 0} \frac{\tan(\pi \sin^2 x) + (|x| - \sin(x[x]))^2}{x^2}$$

[JEE(Main)-2019]

- (1) equals π (2) equals 0 (3) equals $\pi + 1$ (4) does not exist

LT0118

13.
$$\lim_{x \rightarrow \pi/4} \frac{\cot^3 x - \tan x}{\cos \left(x + \frac{\pi}{4} \right)}$$
 is :

[JEE(Main)-2019]

- (1) 4 (2) $8\sqrt{2}$ (3) 8 (4) $4\sqrt{2}$

LT0119

14. $\lim_{x \rightarrow 1^-} \frac{\sqrt{\pi} - \sqrt{2\sin^{-1}x}}{\sqrt{1-x}}$ equal to : [JEE(Main)-2019]

- (1) $\frac{1}{\sqrt{2\pi}}$ (2) $\sqrt{\frac{\pi}{2}}$ (3) $\sqrt{\frac{2}{\pi}}$ (4) $\sqrt{\pi}$

LT0120

15. $\lim_{x \rightarrow 0} \frac{\sin^2 x}{\sqrt{2} - \sqrt{1 + \cos x}}$ equals : [JEE(Main)-2019]

- (1) $2\sqrt{2}$ (2) $4\sqrt{2}$ (3) $\sqrt{2}$ (4) 4

LT0121

16. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function satisfying $f'(3) + f'(2) = 0$.

Then $\lim_{x \rightarrow 0} \left(\frac{1 + f(3+x) - f(3)}{1 + f(2-x) - f(2)} \right)^{\frac{1}{x}}$ is equal to [JEE(Main)-2019]

- (1) e^2 (2) e (3) e^{-1} (4) 1

LT0122

17. If $\lim_{x \rightarrow 1} \frac{x^4 - 1}{x - 1} = \lim_{x \rightarrow k} \frac{x^3 - k^3}{x^2 - k^2}$, then k is : [JEE(Main)-2019]

- (1) $\frac{3}{8}$ (2) $\frac{3}{2}$ (3) $\frac{4}{3}$ (4) $\frac{8}{3}$

LT0123

18. If $\lim_{x \rightarrow 1} \frac{x^2 - ax + b}{x - 1} = 5$, then $a + b$ is equal to :- [JEE(Main)-2019]

- (1) -7 (2) -4 (3) 5 (4) 1

LT0124

19. $\lim_{n \rightarrow \infty} \tan \left\{ \sum_{r=1}^n \tan^{-1} \left(\frac{1}{1+r+r^2} \right) \right\}$ is equal to _____. [JEE(Main)-2021]

LT0153

20. If $\lim_{n \rightarrow \infty} \frac{(n+1)^{k-1}}{n^{k+1}} [(nk+1) + (nk+2) + \dots + (nk+n)] = 33 \cdot \lim_{n \rightarrow \infty} \frac{1}{n^{k+1}} \cdot [1^k + 2^k + 3^k + \dots + n^k]$, then the integral value of k is equal to _____. [JEE(Main)-2022]

LT0154

EXERCISE (JA)

1. Let $L = \lim_{x \rightarrow 0} \frac{a - \sqrt{a^2 - x^2} - \frac{x^2}{4}}{x^4}$, $a > 0$. If L is finite, then - [JEE 2009, 4]

(A) $a = 2$ (B) $a = 1$ (C) $L = \frac{1}{64}$ (D) $L = \frac{1}{32}$

LT0125

2. If $\lim_{x \rightarrow 0} \left[1 + x \ln(1 + b^2) \right]^{\frac{1}{x}} = 2b \sin^2 \theta$, $b > 0$ and $\theta \in (-\pi, \pi]$, then the value of θ is- [JEE 2011, 3M, -1M]

(A) $\pm \frac{\pi}{4}$ (B) $\pm \frac{\pi}{3}$ (C) $\pm \frac{\pi}{6}$ (D) $\pm \frac{\pi}{2}$

LT0126

3. If $\lim_{x \rightarrow \infty} \left(\frac{x^2 + x + 1}{x + 1} - ax - b \right) = 4$, then - [JEE 2012, 3M, -1M]

(A) $a = 1, b = 4$ (B) $a = 1, b = -4$
(C) $a = 2, b = -3$ (D) $a = 2, b = 3$

LT0127

4. Let $\alpha(a)$ and $\beta(a)$ be the roots of the equation $(\sqrt[3]{1+a} - 1)x^2 + (\sqrt{1+a} - 1)x + (\sqrt[6]{1+a} - 1) = 0$ where $a > -1$. Then $\lim_{a \rightarrow 0^+} \alpha(a)$ and $\lim_{a \rightarrow 0^+} \beta(a)$ are [JEE 2012, 3M, -1M]

(A) $-\frac{5}{2}$ and 1 (B) $-\frac{1}{2}$ and -1 (C) $-\frac{7}{2}$ and 2 (D) $-\frac{9}{2}$ and 3

LT0128

5. The largest value of the non-negative integer a for which $\lim_{x \rightarrow 1} \left\{ \frac{-ax + \sin(x-1) + a}{x + \sin(x-1) - 1} \right\}^{\frac{1-x}{1-\sqrt{x}}} = \frac{1}{4}$ is

[JEE(Advanced)-2014, 3]

LT0129

6. Let m and n be two positive integers greater than 1. If $\lim_{\alpha \rightarrow 0} \left(\frac{e^{\cos(\alpha^n)} - e}{\alpha^m} \right) = -\left(\frac{e}{2} \right)$ then the value of

$\frac{m}{n}$ is

[JEE 2015, 4M, -0M]

LT0130

7. Let $\alpha, \beta \in \mathbb{R}$ be such that $\lim_{x \rightarrow 0} \frac{x^2 \sin(\beta x)}{\alpha x - \sin x} = 1$. Then $6(\alpha + \beta)$ equals

[JEE(Advanced)-2016, 3(0)]

LT0131

8. Let $f(x) = \frac{1-x(1+|1-x|)}{|1-x|} \cos\left(\frac{1}{1-x}\right)$ for $x \neq 1$. Then

[JEE(Advanced)-2017, 4]

- (A) $\lim_{x \rightarrow 1^+} f(x)$ does not exist (B) $\lim_{x \rightarrow 1^-} f(x)$ does not exist
(C) $\lim_{x \rightarrow 1^-} f(x) = 0$ (D) $\lim_{x \rightarrow 1^+} f(x) = 0$

LT0132

9. For any positive integer n , define $f_n : (0, \infty) \rightarrow \mathbb{R}$ as

$$f_n(x) = \sum_{j=1}^n \tan^{-1} \left(\frac{1}{1+(x+j)(x+j-1)} \right) \text{ for all } x \in (0, \infty).$$

(Here, the inverse trigonometric function $\tan^{-1}x$ assume values in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.)

Then, which of the following statement(s) is (are) TRUE ?

[JEE(Advanced)-2018]

(A) $\sum_{j=1}^5 \tan^2(f_j(0)) = 55$

(B) $\sum_{j=1}^{10} (1 + f'_j(0)) \sec^2(f_j(0)) = 10$

(C) For any fixed positive integer n , $\lim_{x \rightarrow \infty} \tan(f_n(x)) = \frac{1}{n}$

(D) For any fixed positive integer n , $\lim_{x \rightarrow \infty} \sec^2(f_n(x)) = 1$

LT0133

10. Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be a function. We say that f has

PROPERTY 1 if $\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{\sqrt{|h|}}$ exists and is finite, and

PROPERTY 2 if $\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h^2}$ exists and is finite.

Then which of the following options is/are correct ?

[JEE(Advanced)-2019, 4(-1)]

(1) $f(x) = x|x|$ has PROPERTY 2

(2) $f(x) = x^{2/3}$ has PROPERTY 1

(3) $f(x) = \sin x$ has PROPERTY 2

(4) $f(x) = |x|$ has PROPERTY 1

LT0134

11. Let e denote the base of the natural logarithm. The value of the real number a for which the right hand limit

$$\lim_{x \rightarrow 0^+} \frac{(1-x)^{\frac{1}{x}} - e^{-1}}{x^a}$$

is equal to a nonzero real number, is _____.

[JEE(Advanced)-2020]

LT0141

12. The value of the limit

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{4\sqrt{2}(\sin 3x + \sin x)}{\left(2 \sin 2x \sin \frac{3x}{2} + \cos \frac{5x}{2}\right) - \left(\sqrt{2} + \sqrt{2} \cos 2x + \cos \frac{3x}{2}\right)}$$

is _____

[JEE(Advanced)-2020]

LT0142

13. Let α be a positive real number. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : (\alpha, \infty) \rightarrow \mathbb{R}$ be the functions defined by

$$f(x) = \sin\left(\frac{\pi x}{12}\right) \text{ and } g(x) = \frac{2 \log_e(\sqrt{x} - \sqrt{\alpha})}{\log_e(e^{\sqrt{x}} - e^{\sqrt{\alpha}})}.$$

LT0155

Then the value of $\lim_{x \rightarrow \alpha^+} f(g(x))$ is _____.

[JEE(Advanced)-2022]

14. If $\beta = \lim_{x \rightarrow 0} \frac{e^{x^3} - (1-x^3)^{\frac{1}{3}} + \left((1-x^2)^{\frac{1}{2}} - 1\right) \sin x}{x \sin^2 x}$, then the value of 6β is _____.

LT0156

[JEE(Advanced)-2022]

ANSWER KEY

Do yourself - 1

1. (a) T (b) F (c) F (d) T (e) T (f) T (g) T (h) T (i) F (j) T

Do yourself - 2

1. $-\frac{1}{3}$ 2. 9 3. 2 4. 3 5. 2 6. $\frac{1}{2}$ 7. (A)
8. 10 9. 3a 10. (A) 11. (C) 12. 5050

Do yourself - 3

1. $\frac{\sqrt{q}}{\sqrt{p}}$ 2. $\frac{2}{3\sqrt{3}}$ 3. $\frac{1}{\sqrt{24}}$ 4. (C) 5. (D)

Do yourself - 4

1. 1 2. $-\frac{1}{2}$ 3. 27 4. DNE 5. $\frac{1}{2}$ 6. $\frac{1}{6}$ 7. 3
8. (A) 9. (B)

Do yourself - 5

1. $\frac{\alpha}{\beta}$ 2. $\frac{\sin 2y}{2y}$ 3. $2a \sin a + a^2 \cos a$ 4. -2 5. (C) 6. (A)
7. $\frac{1}{4}$ 8. $\frac{25}{6}$ 9. 2 10. 0.25 11. 2.00

Do yourself - 6

1. e^a 2. $2 \ln 2$ 3. -6 4. $a^a \ln a$ 5. $\frac{3}{2}$ 6. 1 7. 1
8. 0

Do yourself - 7

1. a 2. e^p 3. $e^{\frac{1}{2}}$ 4. e^5

Do yourself - 8

1. 0 2. 5 3. 0 4. e^4 5. e^{-1} 6. e 7. 1
8. $e^{a(x_1 - x_2)}$ 9. $(abc)^{2/3}$ 10. (a) e^{-2} (b) $e^{1/\pi}$ 11. 8.00

Do yourself - 9

1. $\frac{1}{6}$ 2. $\frac{1}{3}$ 3. $-\frac{1}{6}$ 4. 1 5. $-\frac{1}{2}$ 6. $\frac{1}{120}$ 7. (B)
8. (C) 9. $\frac{1}{2}$ 10. (C)

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	B	C	B	A	A	C	B	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	A	D	A	A	C	D	D	C	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	A	A	C	A	D	B	D	C	D

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B,C	A,C	A,B,C	A,D	B,C,D	A,B,C,D	B,C,D	B,C,D	A,B,D	A,C,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,C	B,C	B,D	B,C,D	A,B	A,D	A,C	B,C	A,D	A,B,C,D

EXERCISE # O-3

Que.	1	2	3	4	5	6	7			
Ans.	B	A	A	-1.00	-0.33	0.00	B			
Q.8	A	B	C	D	Q.9	A	B	C	D	
	P,R	P,R	Q	Q		R	S	P	Q	
Q.10	A	B	C	D						
	P,Q,R,S,T	P,R	P,R	S						

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	1	167	0.50	0.50	0.50	-2.00	-0.50	16.00	-0.50

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	3	1	4	1	4	3	2	1	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	4	3	3	2	4	4	1	1	5

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C	D	B	B	0	2	7	A,C	D	2,4
Que.	11	12	13	14						
Ans.	1.00	8	0.50	5						

EXERCISE (O-1)

Straight Objective Type

1. Let $f(x) = \begin{cases} ax+1 & \text{if } x < 1 \\ 3 & \text{if } x = 1 \\ bx^2+1 & \text{if } x > 1 \end{cases}$. If $f(x)$ is continuous at $x = 1$ then $(a - b)$ is equal to-
- (A) 0 (B) 1 (C) 2 (D) 4

CY0001

2. For the function $f(x) = \begin{cases} \frac{1}{x+2\left(\frac{1}{x-2}\right)}, & x \neq 2 \\ k, & x = 2 \end{cases}$ which of the following holds?
- (A) $k = 1/2$ and f is continuous at $x = 2$ (B) $k \neq 0, 1/2$ and f is continuous at $x = 2$
 (C) f can not be continuous at $x = 2$ (D) $k = 0$ and f is continuous at $x = 2$.

CY0002

3. The function $f(x) = \frac{4-x^2}{4x-x^3}$, is -
- (A) discontinuous at only one point in its domain.
 (B) discontinuous at two points in its domain.
 (C) discontinuous at three points in its domain.
 (D) continuous everywhere in its domain.

CY0003

4. If $f(x) = \begin{cases} -4\sin x + \cos x & \text{for } x \leq -\frac{\pi}{2} \\ a\sin x + b & \text{for } -\frac{\pi}{2} < x < \frac{\pi}{2} \\ \cos x + 2 & \text{for } x \geq \frac{\pi}{2} \end{cases}$ is continuous then :
- (A) $a = -1, b = 3$ (B) $a = 1, b = -3$
 (C) $a = 1, b = 3$ (D) $a = -1, b = -3$

CY0004

5. The function $f(x) = \begin{cases} \frac{1}{4}(3x^2+1) & -\infty < x \leq 1 \\ 5-4x & 1 < x < 4 \\ 4-x & 4 \leq x < \infty \end{cases}$ is -
- (A) continuous at $x = 1$ & $x = 4$
 (B) continuous at $x = 1$, discontinuous at $x = 4$
 (C) continuous at $x = 4$, discontinuous at $x = 1$
 (D) discontinuous at $x = 1$ & $x = 4$

CY0005

6. If $f(x) = \frac{x^2 - bx + 25}{x^2 - 7x + 10}$ for $x \neq 5$ and f is continuous at $x = 5$, then $f(5)$ has the value equal to-
- (A) 0 (B) 5 (C) 10 (D) 25

CY0006

7. If $f(x) = \frac{x - e^x + \cos 2x}{x^2}$, $x \neq 0$ is continuous at $x = 0$, then -

- (A) $f(0) = \frac{5}{2}$ (B) $[f(0)] = -2$ (C) $\{f(0)\} = -0.5$ (D) $[f(0)]. \{f(0)\} = -1.5$

where $[.]$ and $\{.\}$ denotes greatest integer and fractional part function

CY0007

8. $y = f(x)$ is a continuous function such that its graph passes through $(a, 0)$. Then $\lim_{x \rightarrow a} \frac{\log_e(1 + 3f(x))}{2f(x)}$ is-
- (A) 1 (B) 0 (C) $\frac{3}{2}$ (D) $\frac{2}{3}$

CY0008

9. In $[1, 3]$, the function $[x^2 + 1], [.]$ denoting the greatest integer function, is continuous -
- (A) for all x (B) for all x except at nine points
- (C) for all x except at seven points (D) for all x except at eight points

CY0009

10. Number of points of discontinuity of $f(x) = [2x^3 - 5]$ in $[1, 2)$, is equal to- (where $[x]$ denotes greatest integer less than or equal to x)
- (A) 14 (B) 13 (C) 10 (D) 8

CY0010

11. Given $f(x) = \begin{cases} |x+1| & \text{if } x < -2 \\ 2x+3 & \text{if } -2 \leq x < 0 \\ x^2+3 & \text{if } 0 \leq x < 3 \\ x^3-15 & \text{if } x \geq 3 \end{cases}$. Then number of point(s) of discontinuity of $f(x)$ is-
- (A) 0 (B) 1 (C) 2 (D) 3

CY0011

12. If $f(x)$ is continuous and $f\left(\frac{9}{2}\right) = \frac{2}{9}$, then the value of $\lim_{x \rightarrow 0} f\left(\frac{1 - \cos 3x}{x^2}\right)$ is-
- (A) $\frac{2}{9}$ (B) $\frac{9}{2}$ (C) 0 (D) data insufficient

CY0012

13. f is a continuous function on the real line. Given that $x^2 + (f(x) - 2)x - \sqrt{3} \cdot f(x) + 2\sqrt{3} - 3 = 0$.

Then the value of $f(\sqrt{3})$

- (A) can not be determined (B) is $2(1 - \sqrt{3})$
(C) is zero (D) is $\frac{2(\sqrt{3} - 2)}{\sqrt{3}}$

CY0013

14. The function $f(x) = [x]^2 - [x^2]$ (where $[y]$ is the greatest integer less than or equal to y), is discontinuous at :

- (A) all integers (B) all integers except 0 & 1
(C) all integers except 0 (D) all integers except 1

CY0014

15. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function $\forall x \in \mathbb{R}$ and $f(x) = 5 \forall x \in \text{irrational}$. Then the value of $f(3)$ is -

- (A) 1 (B) 2 (C) 5 (D) cannot determine

CY0015

16. The function $f(x) = [x] \cdot \cos \frac{2x-1}{2} \pi$, where $[\cdot]$ denotes the greatest integer function, is discontinuous at :-

- (A) all x (B) all integer points (C) no x (D) x which is not an integer

CY0017

17. If $f(x) = \begin{cases} \frac{x^2 + ax + b}{x^2 - 3x + 2} & x \neq 1 \\ c & x = 1 \end{cases}$ is continuous at $x = 1$, then $(2a + b + c)$ is-

CY0067

- (A) 0 (B) 1 (C) 3 (D) -3

18. Number of points where $f(x) = |x - \text{sgn}(x)|$ is non differentiable, is ($\text{sgn}(\cdot)$ denotes signum function)

- (A) 0 (B) 1 (C) 2 (D) 3

CY0068

19. Let $f : \mathbb{R} \rightarrow \mathbb{Q}$ be a continuous function such that $f(2) = 7$, then-

CY0069

- (A) $f(x)$ is always an even function
(B) $f(x)$ is always an odd function
(C) nothing can said about $f(x)$ being even or odd
(D) $f(x)$ is an increasing function

20. Consider $f(x) = \begin{cases} \frac{\sin(2\pi \sec x)}{e^x - 1 - x} & x \neq 0 \text{ or } (2n+1)\frac{\pi}{2}, n \in \mathbb{I} \\ k & x = 0 \end{cases}$

CY0070

Value of k for which $f(x)$ is continuous at $x = 0$, is -

- (A) $\frac{\pi}{2}$ (B) π (C) 2π (D) 4π

21. Let $f(x) = \begin{cases} \frac{\ln \cos x}{ax} & , x > 0 \\ 0 & , x = 0 \\ \frac{e^{x^2} - 1}{bx} & , x < 0 \end{cases}$. If $f'(0) = \frac{1}{4}$, then CY0071

- (A) $a + b = 2$ (B) $b - a = 2$ (C) $a + b = -2$ (D) $b - a = 4$

22. If the function $f(x) = \begin{cases} 2x^2 - 3 & \text{if } 0 < x \leq 1 \\ x^2 + bx - 1 & \text{if } 1 < x < 2 \end{cases}$ is continuous at every point of its domain, then the value of b is- CY0072

- (A) -1 (B) 0 (C) 1 (D) 2

23. Let $f(x) = \begin{cases} \frac{e^x - e^{\sin x}}{ax^3} & ; x < 0 \\ b & ; x = 0 \\ \frac{x}{\ln(1+4x)} & ; x > 0 \end{cases}$. If f is continuous at $x = 0$, then $(3a + 4b)$ is equal to - CY0073

- (A) -3 (B) 0 (C) 3 (D) 4

24. Let $f(x) = \begin{cases} \frac{\tan(bx^3)}{x^3} & , x \neq 0 \\ \frac{5}{4} + \frac{3}{4b} & , x = 0 \end{cases}$, be continuous at $x = 0$, then CY0074

- (A) $b = \frac{5 + \sqrt{73}}{8}$ (B) $b = \frac{3 + \sqrt{73}}{8}$ (C) $b = \frac{-5 + \sqrt{73}}{8}$ (D) $b = \frac{-3 + \sqrt{73}}{8}$

25. The number of values of x , $x \in [-2, 3]$ where $f(x) = [x^2] \sin(\pi x)$ is discontinuous is CY0075
(where $[.]$ denotes greatest integer function)-

- (A) 19 (B) 8 (C) 6 (D) 0

EXERCISE (O-2)

Multiple Correct Answer Type

1. If $f(x) = \begin{cases} \sin^{-1}\left(\frac{a}{\ell \ln x} - \frac{b}{x-1}\right) & x \neq 1 \\ \frac{\pi}{3} & x = 1 \end{cases}$ is continuous at $x = 1$, then- CY0084

(A) $a = b$ (B) $a = \sqrt{3}$ (C) $b = \sqrt{3}$ (D) $b = -\sqrt{3}$

2. If $f(x) = \text{sgn}(x^5)$, then which of the following is/are **false** (where sgn denotes signum function) - CY0085
- (A) $f'(0^+) = 1$
 (B) $f'(0^-) = -1$
 (C) f is continuous but not differentiable at $x = 0$
 (D) f is discontinuous at $x = 0$

3. Let $f(x) = \begin{cases} \frac{\ell \ln(1+x) - x + kx^2}{x^3} & ; x > 0 \\ \ell & ; x = 0 \\ \frac{\tan \frac{x}{3}}{x} & ; x < 0 \end{cases}$, then- CY0086

(A) $f(x)$ is continuous at $x = 0$ if $k = \frac{1}{2}, \ell = \frac{1}{3}$
 (B) $f(x)$ is discontinuous at $x = 0$ if $\ell \neq \frac{1}{3}$
 (C) $f(x)$ has irremovable discontinuity at $x = 0$ if $k \neq \frac{1}{2}$
 (D) $f(x)$ has removable discontinuity at $x = 0$ if $k = \frac{1}{2}, \ell \neq \frac{1}{3}$

4. Which of the following function(s) is/are discontinuous at $x = 0$?

(A) $f(x) = \sin \frac{\pi}{2x}, x \neq 0$ and $f(0) = 1$ (B) $g(x) = x \sin\left(\frac{\pi}{x}\right), x \neq 0$ and $g(0) = \pi$
 (C) $h(x) = \frac{|x|}{x}, x \neq 0$ and $h(0) = 1$ (D) $k(x) = \frac{1}{1+e^{\cot x}}, x \neq 0$ and $k(0) = 0$.

CY0018

5. A function $f(x)$ is defined as $f(x) = \frac{A \sin x + \sin 2x}{x^3}, (x \neq 0)$. If the function is continuous at $x = 0$, then -
- (A) $A = -2$ (B) $f(0) = -1$ (C) $A = 1$ (D) $f(0) = 1$

CY0019

6. Which of the following function(s) can't be defined at $x = 0$ to make it continuous at $x = 0$?

(A) $f(x) = \frac{1}{1 + 2^{\frac{1}{x}}}$ (B) $f(x) = \tan^{-1} \frac{1}{x}$ (C) $f(x) = \frac{e^{\frac{1}{x}} - 1}{\frac{1}{e^x} + 1}$ (D) $f(x) = \frac{1}{\ln |x|}$

CY0020

7. Which of the following function(s) can be defined continuously at $x = 0$?

(A) $f(x) = \frac{1}{1 + 2^{\cot x}}$ (B) $f(x) = \cos\left(\frac{|\sin x|}{x}\right)$
 (C) $f(x) = x \sin \frac{\pi}{x}$ (D) $f(x) = \frac{1}{\ln |x|}$

CY0021

8. Which of the following function(s) can be defined continuously at $x = 0$?

(A) $f(x) = \frac{1 - \sec^2 2x}{4x^2}$ (B) $g(x) = \frac{\csc x - 1}{x \csc x}$ (where $\csc x = \operatorname{cosec} x$)
 (C) $h(x) = \frac{\sin 5x}{x}$ (D) $l(x) = (1 + 2x^2)^{\frac{1}{x^2}}$

CY0022

9. If f is defined on an interval $[a, b]$. Which of the following statement(s) is/are INCORRECT ?

- (A) If $f(a)$ and $f(b)$, have opposite sign, then there must be a point $c \in (a, b)$ such that $f(c) = 0$.
 (B) If f is continuous on $[a, b]$, $f(a) < 0$ and $f(b) > 0$, then there must be a point $c \in (a, b)$ such that $f(c) = 0$.
 (C) If f is continuous on $[a, b]$ and there is a point c in (a, b) such that $f(c) = 0$, then $f(a)$ and $f(b)$ have opposite sign.
 (D) If f has no zeroes on $[a, b]$, then $f(a)$ and $f(b)$ have the same sign.

CY0023

10. Which of the following functions can be defined at indicated point so that resulting function is continuous -

(A) $f(x) = \frac{x^2 - 2x - 8}{x + 2}$ at $x = -2$ (B) $f(x) = \frac{x - 7}{|x - 7|}$ at $x = 7$
 (C) $f(x) = \frac{x^3 + 64}{x + 4}$ at $x = -4$ (D) $f(x) = \frac{3 - \sqrt{x}}{9 - x}$ at $x = 9$

CY0024

11. In which of the following cases the given equations has atleast one root in the indicated interval ?

- (A) $x - \cos x = 0$ in $(0, \pi/2)$
 (B) $x + \sin x = 1$ in $(0, \pi/6)$
 (C) $\frac{a}{x-1} + \frac{b}{x-3} = 0$, $a, b > 0$ in $(1, 3)$
 (D) $f(x) - g(x) = 0$ in (a, b) where f and g are continuous on $[a, b]$ and $f(a) > g(a)$ and $f(b) < g(b)$.

CY0025

12. Given $f(x) = \sum_{r=1}^n \tan\left(\frac{x}{2^r}\right) \sec\left(\frac{x}{2^{r-1}}\right); r, n \in \mathbb{N}$

$$g(x) = \lim_{n \rightarrow \infty} \frac{\ln\left(f(x) + \tan \frac{x}{2^n}\right) - \left(f(x) + \tan \frac{x}{2^n}\right)^n \cdot \left[\sin\left(\tan \frac{x}{2}\right)\right]}{1 + \left(f(x) + \tan \frac{x}{2^n}\right)^n}$$

$= k$ for $x = \frac{\pi}{4}$ and the domain of $g(x)$ is $(0, \pi/2)$.

where $[]$ denotes the greatest integer function.

If $g(x)$ is continuous at $x = \pi/4$, then

(A) $k = 0$

(B) $k = 1$

(C) $g(x) = \begin{cases} \ln(\tan x) & \text{if } 0 < x < \frac{\pi}{4} \\ 0 & \text{if } \frac{\pi}{4} \leq x < \frac{\pi}{2} \end{cases}$

(D) $g(x)$ is continuous everywhere

CY0087

13. Let $f(x) = x^3 - x^2 - 3x - 1$ and $h(x) = \frac{f(x)}{g(x)}$, where h is a rational function such that

(a) Domain of $h(x)$ is $\mathbb{R} - \{-1\}$

(b) $\lim_{x \rightarrow \infty} h(x) = \infty$ and (c) $\lim_{x \rightarrow -1} h(x) = \frac{1}{2}$.

Which of the following is/are True?

(A) $\lim_{x \rightarrow 0} (3h(x) + f(x) - 2g(x)) = -\frac{39}{4}$

(B) $g(x) = 4(x + 1)$

(C) $\lim_{x \rightarrow 0} (3h(x) + f(x) - 2g(x)) = \frac{39}{4}$

(D) $g(x) = 2(x + 1)$

CY0088

14. If the function $f(x) = \frac{3x^2 + ax + a + 3}{x^2 + x - 2}$, ($x \neq -2$) is continuous at $x = -2$, then

CY0089

(A) $a = 15$

(B) $a = -15$

(C) $f(-2) = 1$

(D) $f(-2) = -1$

15. Let $f(x) = \begin{cases} |ax + 3| & \text{if } x \leq -1 \\ |3x + a| & \text{if } -1 < x \leq 0 \\ \frac{b \sin 2x}{x} - 2b & \text{if } 0 < x < \pi \\ \cos^2 x - 3 & \text{if } x \geq \pi \end{cases}$. If $f(x)$ is continuous for all $x \in \mathbb{R}$, then

CY0090

(A) $a = 0$

(B) $a = 1$

(C) $b = 1$

(D) $b = -1$

16. Let $f(x) = \begin{cases} \frac{\ln \cos x}{\sqrt[4]{1+x^2}-1} & \text{if } x > 0 \\ \frac{e^{\sin 4x}-1}{\ln(1+\tan 2x)} & \text{if } x < 0 \end{cases}$ CY0091

- (A) $f(0^+) = -2$ (B) $f(0^-) = 2$ (C) $f(0^+) = 2$ (D) $f(0)$ can not be defined

17. Let $f(x) = \begin{cases} \frac{\left(\frac{\pi}{2} - \sin^{-1}(1-\{x\}^2)\right) \sin^{-1}(1-\{x\})}{\sqrt{2}(\{x\}-\{x\}^3)} & \text{for } x \neq 0 \\ \frac{\pi}{2} & \text{for } x = 0 \end{cases}$ where $\{x\}$ is the fractional part of x .

Consider another function $g(x)$; such that

CY0092

$$g(x) = \begin{cases} f(x) & \text{for } x \geq 0 \\ 2\sqrt{2}f(x) & \text{for } x < 0 \end{cases}$$

(A) $f(0^-) = \frac{\pi}{4\sqrt{2}}$

(B) f is discontinuous at $x = 0$

(C) $g(0^+) = \pi/2$

(D) g is discontinuous at $x = 0$

18. Let the function $f(x) = \begin{cases} ax-b & \text{for } x \leq 1 \\ 3x & \text{for } 1 < x < 2 \\ bx^2-a & \text{for } x \geq 2 \end{cases}$ is continuous at $x = 1$ but discontinuous at $x = 2$, then

(A) $a - b = 3$

(B) $a - b = 6$

(C) $b \neq 3$

(D) $b = 3$

CY0093

19. A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined as $f(x) = \lim_{n \rightarrow \infty} \frac{ax^2 + bx + c + e^{nx}}{1 + c \cdot e^{nx}}$ where f is continuous on $\mathbb{R}$, then

(A) $a \in \mathbb{R}$

(B) $b \in \mathbb{R}$

(C) $c = -1$

(D) $c = 1$

CY0094

20. Suppose that $f(x) = x^3 - 3x^2 - 4x + 12$ and $h(x) = \begin{cases} \frac{f(x)}{x-3} & , \quad x \neq 3 \\ K & , \quad x = 3 \end{cases}$ then

CY0095

(A) $f(x) = 0$ has 3 real roots

(B) If $K = 5$ then $h(x)$ is continuous at $x = 3$

(C) for $K = 5$, $h(x)$ is an even function

(D) for $K = 5$, $h(x)$ is an odd function

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 and 2

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as,

$$f(x) = \begin{cases} 1 - |x| & , |x| \leq 1 \\ 0 & , |x| > 1 \end{cases} \text{ and } g(x) = f(x-1) + f(x+1), \forall x \in \mathbb{R}. \text{ Then}$$

1. The value of $g(x)$ is :

CY0076

$$(A) g(x) = \begin{cases} 0 & , x \leq -3 \\ 2+x & , -3 \leq x \leq -1 \\ -x & , -1 < x \leq 0 \\ x & , 0 < x \leq 1 \\ 2-x & , 1 < x \leq 3 \\ 0 & , x > 3 \end{cases}$$

$$(B) g(x) = \begin{cases} 0 & , x \leq -2 \\ 2+x & , -2 \leq x \leq -1 \\ -x & , -1 < x \leq 0 \\ x & , 0 < x \leq 1 \\ 2-x & , 1 < x \leq 2 \\ 0 & , x > 2 \end{cases}$$

$$(C) g(x) = \begin{cases} 0 & , x \leq 0 \\ 2+x & , 0 < x < 1 \\ -x & , 1 \leq x \leq 2 \\ x & , 2 < x < 3 \\ 2-x & , 3 \leq x < 4 \\ 0 & , 4 \leq x \end{cases}$$

(D) none of these

2. The function $g(x)$ is continuous for, $x \in$

CY0077

(A) $\mathbb{R} - \{0, 1, 2, 3, 4\}$

(B) $\mathbb{R} - \{-2, -1, 0, 1, 2\}$

(C) $\mathbb{R}$

(D) none of these

Paragraph for Question 3 to 5

$$\text{Let } f(x) = \lim_{n \rightarrow \infty} \frac{\ln x + x^n}{(x-1) + x^n} \quad (x > 0).$$

On the basis of above information, answer the following questions :

3. $\lim_{x \rightarrow 1^-} \frac{\sin(f(x)-1)}{(x-1)}$ is equal to -

CY0078

(A) 0

(B) $\frac{1}{2}$

(C) $-\frac{1}{2}$

(D) 1

4. For $f(x)$ which of the following statement is true -

CY0079

(A) $f(x)$ is discontinuous at $x = 1$

(B) non-negative difference between LHL & RHL at $x = 1$ is equal to 2.

(C) $f(x)$ is many-one function

(D) $f(x)$ is one-one function

5. The number of solutions of the equation $f(x) = \frac{1}{(x-1)^2}$ is -

CY0080

(A) 0

(B) 1

(C) 2

(D) 3

Paragraph for Question 6 and 7

Consider $f(x) = \frac{1}{\ln|x|}$ and $g(x) = \tan^{-1}|x|$.

On the basis of above information, answer the following questions :

6. Sum of solutions of the equation $\frac{g(x)}{f(x)} = \frac{1}{2}$ will be less than or equal to : **CY0081**
 (A) 0 (B) 1 (C) 2 (D) 3
7. The value(s) of x at which $h(x) = f(x) + g(x)$ will be discontinuous is/are : **CY0082**
 (A) 0 (B) 1 (C) -1 (D) 2

Matrix Match Type

8. Match column-I with column-II **CY0083**

Column I		Column II	
(P)	The number of points of discontinuity of $f(x) = ([x] + [-x])x^2$ in $(-5, 5)$ is equal to, (where $[\cdot]$ is greatest integer function)	(1)	0
(Q)	The number of elements in the range of $f(x) = \sqrt{\cos(\cos x)} + \sin^{-1}\left(\frac{1+x^2}{2x}\right)$ is equal to	(2)	2
(R)	The number of integral values of x for which $ 3x-1 < 2x-1 + x $	(3)	8
(S)	The number of points where $f(x) = \operatorname{sgn}(\sin x) + \operatorname{sgn}(x^2 - \pi x)$ is discontinuous in $(0, 2\pi)$ is equal to (where sgn indicates signum function)	(4)	1

- (A) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 4$ (B) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 4$
 (C) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 1$ (D) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 1; S \rightarrow 2$

9. **Column-I** **Column-II**

- (A) $\lim_{x \rightarrow 1} \frac{x^3 - 1}{\ln x}$ is (P) 2
- (B) $\lim_{x \rightarrow 0} \frac{x(\cos x - \cos 2x)}{2 \sin x - \sin 2x}$ is (Q) 3
- (C) $\lim_{x \rightarrow 0} \frac{\tan x \sqrt{\tan x} - \sin x \sqrt{\sin x}}{x^3 \cdot \sqrt{x}}$ is (R) $\frac{3}{2}$
- (D) If $f(x) = \cos(x \cos \frac{1}{x})$ and $g(x) = \frac{\ln(\sec^2 x)}{x \sin x}$ are (S) $\frac{3}{4}$
 both continuous at $x = 0$ then $f(0) + g(0)$ equals

CY0026

10. Match the function in column-I with its behaviour at $x = 0$ in column-II, where $[.]$ denotes greatest integer function & $\text{sgn}(x)$ denotes signum function.

Column-I

- (A) $f(x) = [x][1 + x]$
 (B) $f(x) = [-x][1 + x]$
 (C) $f(x) = (\text{sgn}(x))[2 - x][1 + |x|]$
 (D) $f(x) = [\cos x]$

Column-II

- (P) LHL exist at $x = 0$
 (Q) RHL exist at $x = 0$
 (R) Continuous at $x = 0$
 (S) $\lim_{x \rightarrow 0} f(x)$ exists but function is discontinuous at $x = 0$
 (T) $\lim_{x \rightarrow 0} f(x)$ does not exist

CY0027

EXERCISE (O-4)

Numerical Grid Type

1. The function $f(x) = \begin{cases} \left(\frac{6}{5}\right)^{\frac{\tan 6x}{\tan 5x}} & \text{if } 0 < x < \frac{\pi}{2} \\ b + 2 & \text{if } x = \frac{\pi}{2} \\ (1 + |\cos x|)^{\left(\frac{a|\tan x|}{b}\right)} & \text{if } \frac{\pi}{2} < x < \pi \end{cases}$

CY0096

If $f(x)$ is continuous at $x = \pi/2$, then the value of $(a - b)$ is

2. Let $f(x) = \begin{cases} \frac{1 - \sin \pi x}{1 + \cos 2\pi x}, & x < \frac{1}{2} \\ p, & x = \frac{1}{2} \\ \frac{\sqrt{2x-1}}{\sqrt{4+\sqrt{2x-1}}-2}, & x > \frac{1}{2} \end{cases}$. The number of all possible values of 'p' so that the function

$f(x)$ is continuous at $x = 1/2$, is -

CY0097

3. Let $f(x) = \begin{cases} 1+x, & 0 \leq x \leq 2 \\ 3-x, & 2 < x \leq 3 \end{cases}$ and $g(x) = f[f(x)]$ & hence the number of points of discontinuity of $g(x)$ is

CY0098

4. If $f(x)$ is continuous at $x = \frac{\pi}{2}$ where $f(x) = \begin{cases} \frac{1 - \sin^3 x}{3\cos^2 x} & \text{if } x < \frac{\pi}{2} \\ a & \text{if } x = \frac{\pi}{2} \\ \frac{b(1 - \sin x)}{(\pi - 2x)^2} & \text{if } x > \frac{\pi}{2} \end{cases}$,

CY0099

then the value of $(4a + 3b)$ is

5. If the function $f(x) = \begin{cases} \frac{\sin(a+1)x + \sin x}{x} & \text{for } x < 0 \\ c & \text{for } x = 0 \\ \frac{(x+bx^2)^{1/2} - x^{1/2}}{bx^{3/2}} & \text{for } x > 0 \end{cases}$ is continuous at $x = 0$, then the value of $(2a + 14c)$ is

CY0100

6. If $f(x) = \frac{\sin 3x + A \sin 2x + B \sin x}{x^5}$ ($x \neq 0$) is continuous at $x = 0$, then the value of $(A + B + f(0))$ is

CY0101

7. If $f(x) = x + \{-x\} + [x]$, where $[x]$ is the integral part & $\{x\}$ is the fractional part of x , then the number of points of discontinuity of $f(x)$ in $[-2, 2]$ is

CY0102

8. Let $f(x) = \begin{cases} (\sin x + \cos x)^{\cos e c x} & ; \quad -\frac{\pi}{2} < x < 0 \\ a & ; \quad x = 0 \\ \frac{e^{1/x} + e^{2/x} + e^{3/|x|}}{ae^{2/x} + be^{3/|x|}} & ; \quad 0 < x < \frac{\pi}{2} \end{cases}$

CY0103

If $f(x)$ is continuous at $x = 0$, then the value of $(a^2 \cdot b^2)$ is

9. Let f be a real valued continuous function on $\mathbb{R}$ and satisfying $f(-x) - f(x) = 0 \forall x \in \mathbb{R}$. If $f(-5) = 5$, $f(-2) = 4$, $f(3) = -2$ and $f(0) = 0$ then find the minimum number of zero's of the equation $f(x) = 0$.

CY0046

10. Find the number of points of discontinuity of the function $f(x) = [5x] + \{3x\}$ in $[0, 5]$ where $[y]$ and $\{y\}$ denote largest integer less than or equal to y and fractional part of y respectively.

CY0047

EXERCISE (JM)

1. The function $f : \mathbb{R}/\{0\} \rightarrow \mathbb{R}$ given by $f(x) = \frac{1}{x} - \frac{2}{e^{2x} - 1}$ can be made continuous at

$x = 0$ by defining $f(0)$ as-

[AIEEE 2007]

- (1) 2 (2) -1 (3) 0 (4) 1

CY0051

2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function defined by $f(x) = \frac{1}{e^x + 2e^{-x}}$

Statement-1 : $f(c) = \frac{1}{3}$, for some $c \in \mathbb{R}$.

Statement-2 : $0 < f(x) \leq \frac{1}{2\sqrt{2}}$, for all $x \in \mathbb{R}$.

[AIEEE-2010]

- (1) Statement-1 is true, Statement-2 is true ; Statement-2 is a correct explanation for Statement-1.
 (2) Statement-1 is true, Statement-2 is true ; Statement-2 is not a correct explanation for statement-1.
 (3) Statement-1 is true, Statement-2 is false.
 (4) Statement-1 is false, Statement-2 is true.

CY0052

3. The values of p and q for which the function $f(x) = \begin{cases} \frac{\sin(p+1)x + \sin x}{x} & , x < 0 \\ q & , x = 0 \\ \frac{\sqrt{x+x^2} - \sqrt{x}}{x^{\frac{3}{2}}} & , x > 0 \end{cases}$ is continuous for all x in $\mathbb{R}$,

are :-

[AIEEE 2011]

- (1) $p = -\frac{3}{2}, q = \frac{1}{2}$ (2) $p = \frac{1}{2}, q = \frac{3}{2}$ (3) $p = \frac{1}{2}, q = -\frac{3}{2}$ (4) $p = \frac{5}{2}, q = \frac{1}{2}$

CY0053

4. Define $F(x)$ as the product of two real functions $f_1(x) = x, x \in \mathbb{R}$,

and $f_2(x) = \begin{cases} \sin \frac{1}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$ as follows : $F(x) = \begin{cases} f_1(x) \cdot f_2(x) & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$

[AIEEE 2011]

Statement-1 : $F(x)$ is continuous on $\mathbb{R}$.

Statement-2 : $f_1(x)$ and $f_2(x)$ are continuous on $\mathbb{R}$.

- (1) Statement-1 is false, statement-2 is true.
 (2) Statement-1 is true, statement-2 is true; Statement-2 is correct explanation for statement 1.
 (3) Statement-1 is true, statement-2 is true, statement-2 is not a correct explanation for statement 1
 (4) Statement-1 is true, statement-2 is false

CY0054

5. If $f(x)$ is continuous and $f(9/2) = 2/9$, then $\lim_{x \rightarrow 0} f\left(\frac{1 - \cos 3x}{x^2}\right)$ is equal to : [JEE Mains Offline-2014]
 (1) $9/2$ (2) 0 (3) $2/9$ (4) $8/9$

CY0055

6. If the function $f(x) = \begin{cases} \frac{\sqrt{2 + \cos x} - 1}{(\pi - x)^2}, & x \neq \pi \\ k, & x = \pi \end{cases}$ is continuous at $x = \pi$, then k equals:-

[JEE Mains Offline-2014]

- (1) $\frac{1}{4}$ (2) $\frac{1}{2}$ (3) 2 (4) 0

CY0056

7. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as : $f(x) = \begin{cases} 5, & \text{if } x \leq 1 \\ a + bx, & \text{if } 1 < x < 3 \\ b + 5x, & \text{if } 3 \leq x < 5 \\ 30, & \text{if } x \geq 5 \end{cases}$. Then, f is :

[JEE(Main)-2019]

- (1) continuous if $a = 5$ and $b = 5$ (2) continuous if $a = -5$ and $b = 10$
 (3) continuous if $a = 0$ and $b = 5$ (4) not continuous for any values of a and b

CY0057

8. Let $f : [-1, 3] \rightarrow \mathbb{R}$ be defined as $f(x) = \begin{cases} |x| + [x], & -1 \leq x < 1 \\ x + |x|, & 1 \leq x < 2 \\ x + [x], & 2 \leq x \leq 3 \end{cases}$,

where $[t]$ denotes the greatest integer less than or equal to t . Then, f is discontinuous at:

[JEE(Main)-2019]

- (1) four or more points (2) only one point
 (3) only two points (4) only three points

CY0058

9. If the function f defined on $\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$ by $f(x) = \begin{cases} \frac{\sqrt{2} \cos x - 1}{\cot x - 1}, & x \neq \frac{\pi}{4} \\ k, & x = \frac{\pi}{4} \end{cases}$ is continuous, then k is equal to

[JEE(Main)-2019]

- (1) $\frac{1}{2}$ (2) 1 (3) $\frac{1}{\sqrt{2}}$ (4) 2

CY0059

10. If $f(x) = [x] - \left\lfloor \frac{x}{4} \right\rfloor$, $x \in \mathbb{R}$, where $[x]$ denotes the greatest integer function, then :

[JEE(Main)-2019]

- (1) Both $\lim_{x \rightarrow 4^-} f(x)$ and $\lim_{x \rightarrow 4^+} f(x)$ exist but are not equal
 (2) $\lim_{x \rightarrow 4^-} f(x)$ exists but $\lim_{x \rightarrow 4^+} f(x)$ does not exist
 (3) $\lim_{x \rightarrow 4^+} f(x)$ exists but $\lim_{x \rightarrow 4^-} f(x)$ does not exist
 (4) f is continuous at $x = 4$

CY0060

11. If the function $f(x) = \begin{cases} a|\pi - x| + 1, & x \leq 5 \\ b|x - \pi| + 3, & x > 5 \end{cases}$ is continuous at $x = 5$, then the value of $a - b$ is :-

[JEE(Main)-2019]

- (1) $\frac{2}{5 - \pi}$ (2) $\frac{2}{\pi - 5}$ (3) $\frac{2}{\pi + 5}$ (4) $\frac{-2}{\pi + 5}$

CY0061

12. If $f(x) = \begin{cases} \frac{\sin(p+1) + \sin x}{x}, & x < 0 \\ q, & x = 0 \\ \frac{\sqrt{x+x^2} - \sqrt{x}}{x^{3/2}}, & x > 0 \end{cases}$ is continuous at $x = 0$, then the ordered pair (p, q) is equal to :

[JEE(Main)-2019]

- (1) $\left(\frac{5}{2}, \frac{1}{2}\right)$ (2) $\left(-\frac{3}{2}, -\frac{1}{2}\right)$ (3) $\left(-\frac{1}{2}, \frac{3}{2}\right)$ (4) $\left(-\frac{3}{2}, \frac{1}{2}\right)$

CY0062

EXERCISE (JA)

1. Discuss the continuity of the function $f(x) = \begin{cases} \frac{e^{1/(x-1)} - 2}{e^{1/(x-1)} + 2}, & x \neq 1 \\ 1, & x = 1 \end{cases}$

at $x = 1$.

[REE 2001 (Mains), 3 out of 100]

CY0063

2. For every integer n , let a_n and b_n be real numbers. Let function $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by

$$f(x) = \begin{cases} a_n + \sin \pi x, & \text{for } x \in [2n, 2n+1] \\ b_n + \cos \pi x, & \text{for } x \in (2n-1, 2n) \end{cases}, \text{ for all integers } n.$$

If f is continuous, then which of the following holds(s) for all n ?

[JEE 2012, 4]

- (A) $a_{n-1} - b_{n-1} = 0$ (B) $a_n - b_n = 1$ (C) $a_n - b_{n+1} = 1$ (D) $a_{n-1} - b_n = -1$

CY0064

3. For every pair of continuous function $f, g : [0, 1] \rightarrow \mathbb{R}$ such that

$$\max\{f(x) : x \in [0, 1]\} = \max\{g(x) : x \in [0, 1]\},$$

the correct statement(s) is(are) :

- (A) $(f(c))^2 + 3f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0, 1]$
 (B) $(f(c))^2 + f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0, 1]$
 (C) $(f(c))^2 + 3f(c) = (g(c))^2 + g(c)$ for some $c \in [0, 1]$
 (D) $(f(c))^2 = (g(c))^2$ for some $c \in [0, 1]$

[JEE(Advanced)-2014, 3]

CY0065

4. Let $[x]$ be the greatest integer less than or equal to x . Then, at which of the following point(s) the function $f(x) = x \cos(\pi(x + [x]))$ is discontinuous?

[JEE(Advanced)-2017, 4(-2)]

- (A) $x = -1$ (B) $x = 0$ (C) $x = 2$ (D) $x = 1$

CY0066

ANSWER KEY

Do yourself - 1

1. 1 2. discontinuous at $x = -2$ 3. $k = \frac{1}{3}$ 4. discontinuous at $x = \frac{\pi}{2}$
 5. continuous at $x = 1$ 6. discontinuous at $x = 2$ 7. $f(0) = a + b$

Do yourself - 2

1. $a = -1$ & $b = 1$ 2. Discontinuous at $x = 1$ & continuous at $x = \frac{\pi}{2}$
 3. Discontinuous at $x = 1$ 5. $A = -1, B = 1$
 6. f is discontinuous at $x = 2$ but continuous for $x \in [-5, 2) \cup (2, 5)$, $g(x)$ is continuous for $x \in [-5, 5)$
 7. continuous everywhere

Do yourself - 3

3. Yes

Do yourself - 4

1. All are discontinuous at $x = 0$.
 2. $f(x)g(x)$ & $f(x) \pm g(x)$ are continuous in $\mathbb{R} - \{x : x = (2n+1)\frac{\pi}{2}; n \in \mathbb{Z}\}$
 $\frac{f(x)}{g(x)}$ is continuous in $\mathbb{R} - \{x : x = \frac{n\pi}{2}; n \in \mathbb{Z}\}$
 3. discontinuous at all integral points 4. discontinuous

Do yourself - 5

1. $x = 0$ 2. $x = \pm \frac{1}{\sqrt{2}}$ 3. continuous nowhere 4. continuous only at $x = 0$

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	C	D	A	B	A	D	C	D	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	A	B	D	C	C	D	D	A	C
Que.	21	22	23	24	25					
Ans.	A	A	C	A	B					

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C	A,B,C	A,B,C,D	A,B,C,D	A,B	A,B,C	B,C,D	A,C,D	A,C,D	A,C,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,B,C,D	A,C,D	A,B	A,D	A,C	A,B,D	A,B,C	A,C	A,B,D	A,B,C

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8		
Ans.	B	C	C	C	C	A,B,C,D	A,B,C	C		
Q.9	A	B	C	D	Q.10	A	B	C	D	
	Q	R	S	P		P,Q,R	P,Q,T	P,Q,T	P,Q,S	

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	0	2	14	4	2.00	5.00	1.00	5.00	30.00

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	1	1	4	3	1	4	4	1	4
Que.	11	12								
Ans.	1	4								

EXERCISE # JEE-ADVANCED

- Discontinuous at $x = 1$; $f(1^+) = 1$ and $f(1^-) = -1$
- B,D
- A,D
- A,C,D

EXERCISE (O-1)

Straight Objective Type

1. Let $f(x) = [\tan^2 x]$, (where $[.]$ denotes greatest integer function). Then -
 (A) $\lim_{x \rightarrow 0} f(x)$ does not exist (B) $f(x)$ is continuous at $x = 0$.
 (C) $f(x)$ is not differentiable at $x = 0$ (D) $f'(0) = 1$
DY0001
2. Let f be differentiable at $x = 0$ and $f'(0) = 1$. Then $\lim_{h \rightarrow 0} \frac{f(h) - f(-2h)}{h} =$
 (A) 3 (B) 2 (C) 1 (D) -1
DY0004
3. Let $g(x) = \begin{cases} 3x^2 - 4\sqrt{x} + 1 & \text{for } x < 1 \\ ax + b & \text{for } x \geq 1 \end{cases}$.
 If $g(x)$ is continuous and differentiable at $x = 1$ then -
 (A) $a = b = 4$ (B) $a = b = -4$
 (C) $a = 4$ and $b = -4$ (D) $a = -4$ and $b = 4$
DY0005
4. If $f(x)f(y) + 2 = f(x) + f(y) + f(xy)$ and $f(1) = 2$, $f'(1) = 2$ then $\text{sgn } f(x)$ is equal to
 (where sgn denotes signum function) -
 (A) 0 (B) 1 (C) -1 (D) 4
DY0006
5. The function $g(x) = \begin{cases} x + b, & x < 0 \\ \cos x, & x \geq 0 \end{cases}$ can be made differentiable at $x = 0$ -
 (A) if b is equal to zero (B) if b is not equal to zero
 (C) if b takes any real value (D) for no value of b
DY0007
6. Which one of the following functions is continuous everywhere in its domain but has atleast one point where it is not differentiable?
 (A) $f(x) = x^{1/3}$ (B) $f(x) = \frac{|x|}{x}$ (C) $f(x) = e^{-x}$ (D) $f(x) = \tan x$
DY0008
7. If the right hand derivative of $f(x) = [x] \tan \pi x$ at $x = 7$ is $k\pi$, then k is equal to
 ($[y]$ denotes greatest integer $\leq y$)
 (A) 6 (B) 7 (C) -7 (D) 49
DY0009
8. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous onto function satisfying $f(x) + f(-x) = 0$, $\forall x \in \mathbb{R}$. If $f(-3) = 2$ and $f(5) = 4$ in $[-5, 5]$, then the equation $f(x) = 0$ has -
 (A) exactly three real roots (B) exactly two real roots
 (C) atleast five real roots (D) atleast three real roots
DY0010

$$9. \text{ Let } f(x) = \begin{cases} \lim_{n \rightarrow \infty} \frac{ax(x-1) \left(\cot \frac{\pi x}{4} \right)^n + (px^2 + 2)}{\left(\cot \frac{\pi x}{4} \right)^n + 1}, & x \in (0,1) \cup (1,2) \\ 0, & x = 1 \end{cases}$$

If $f(x)$ is differentiable for all $x \in (0,2)$ then $(a^2 + p^2)$ equals -

- (A) 18 (B) 20 (C) 22 (D) 24

DY0011

10. If $2x + 3|y| = 4y$, then y as a function of x i.e. $y = f(x)$, is -

- (A) discontinuous at one point
(B) non differentiable at one point
(C) discontinuous & non differentiable at same point
(D) continuous & differentiable everywhere

DY0012

11. If $f(x) = (x^5 + 1)|x^2 - 4x - 5| + \sin|x| + \cos(|x - 1|)$, then $f(x)$ is not differentiable at -

- (A) 2 points (B) 3 points (C) 4 points (D) zero points

DY0013

12. Let $f(x) = \begin{cases} x^3 + 2x^2 & x \in \mathbb{Q} \\ -x^3 + 2x^2 + ax & x \notin \mathbb{Q} \end{cases}$, then the integral value of 'a' so that $f(x)$ is differentiable at

$x = 1$, is

- (A) 2 (B) 6 (C) 7 (D) not possible

DY0014

13. Let $\mathbb{R}$ be the set of real numbers and $f : \mathbb{R} \rightarrow \mathbb{R}$, be a differentiable function such that $|f(x) - f(y)| \leq |x - y|^3 \forall x, y \in \mathbb{R}$. If $f(10) = 100$, then the value of $f(20)$ is equal to -

- (A) 0 (B) 10 (C) 20 (D) 100

DY0015

14. For what triplets of real numbers (a, b, c) with $a \neq 0$ the function

$$f(x) = \begin{cases} x & x \leq 1 \\ ax^2 + bx + c & \text{otherwise} \end{cases} \text{ is differentiable for all real } x?$$

- (A) $\{(a, 1 - 2a, a) \mid a \in \mathbb{R}, a \neq 0\}$ (B) $\{(a, 1 - 2a, c) \mid a, c \in \mathbb{R}, a \neq 0\}$
(C) $\{(a, b, c) \mid a, b, c \in \mathbb{R}, a + b + c = 1\}$ (D) $\{(a, 1 - 2a, 0) \mid a \in \mathbb{R}, a \neq 0\}$

DY0016

15. Number of points of non-differentiability of the function

$g(x) = [x^2]\{\cos^2 4x\} + \{x^2\}[\cos^2 4x] + x^2 \sin^2 4x + [x^2][\cos^2 4x] + \{x^2\}\{\cos^2 4x\}$ in $(-50, 50)$ where $[x]$ and $\{x\}$ denotes the greatest integer function and fractional part function of x respectively, is equal to :-

- (A) 98 (B) 99 (C) 100 (D) 0

DY0017

16. Let $f(x) = [n + p \sin x]$, $x \in (0, \pi)$, $n \in \mathbb{Z}$ and p is a prime number. The number of points where $f(x)$ is not differentiable is :-

Here $[x]$ denotes greatest integer function.

- (A) $p - 1$ (B) $p + 1$ (C) $2p + 1$ (D) $2p - 1$

DY0018

17. The function $f(x) = (x^2 - 1) |x^2 - 3x + 2| + \cos(|x|)$ is NOT differentiable at :

- (A) -1 (B) 0 (C) 1 (D) 2

DY0019

18. Let $f(x) = \begin{cases} ax + bx^2 & ; x < 1 \\ 4ax - 2b + 1 & ; x \geq 1 \end{cases}$ be continuous and differentiable everywhere, then a & b are -

- (A) $\frac{1}{3}, -1$ (B) $\frac{2}{3}, 1$ (C) $\frac{3}{2}, 1$ (D) $\frac{4}{3}, -1$ **DY0078**

19. If $f(x) = \begin{cases} e^{x^2} + x^3 + 1 & ; x > 0 \\ ax^2 + bx + 2 & ; x \leq 0 \end{cases}$ is differentiable at $x = 0$, then -

DY0079

- (A) $a \in \mathbb{R}$ & $b = 0$ (B) $a \in \mathbb{R}$ & $b = 1$ (C) $a \in \mathbb{R}$ & $b = 2$ (D) $a \in \mathbb{R}$ & $b = 3$

20. If $f(x) = [\sin x] + \sqrt{\sin x - [\sin x]}$, where $[.]$ denotes greatest integer function, then $f(x)$ is- **DY0080**

- (A) differentiable at $x = \frac{\pi}{2}$ & $x = \pi$ both (B) differentiable at $x = \frac{\pi}{2}$ but not at $x = \pi$
(C) differentiable at $x = \pi$ but not at $x = \frac{\pi}{2}$ (D) neither differentiable at $x = \pi$ nor at $x = \frac{\pi}{2}$

21. If $f(x) = \begin{cases} \sin^{-1} x + a \cos \pi x & -1 < x < 0 \\ b \cos^{-1} x + \sin \pi x & 0 \leq x < 1 \end{cases}$

DY0081

is differentiable in $(-1, 1)$, then $(2a + b)$ is equal to -

- (A) π^2 (B) $1 + \pi^2$ (C) $1 - \pi^2$ (D) $\pi^2 - 1$

22. If $f(x) = \begin{cases} ae^{ax} & x \leq 0 \\ x^2 + 4x + 2b & x > 0 \end{cases}$

DY0082

& $g(x) = \sin|bx + a - 2|$ both are differentiable at $x = 0$, then $g(x)$ is non differentiable at x is equal to-

- (A) π (B) -4 (C) $\frac{\pi}{4}$ (D) 4

23. An even function $f(x)$, differentiable $\forall x \in \mathbb{R}$ satisfies the condition $f(x - y) = f(x) + f(y) - 4xy - 1$ and $f'(0) = 0$ then $f(x)$ is **DY0083**

- (A) x^2 (B) $x^4 + 1$ (C) $2x^2 + 1$ (D) $2x^4 + 1$

24. Number of points at which the function $f(x) = |\cos|x|| + \cos^{-1} \operatorname{sgn} x + |\ln x|$ is not differentiable in $(0, 2\pi)$ is-

- (A) 1 (B) 2 (C) 3 (D) 4 **DY0084**

25. If $[x]$ denotes the integral part of x and $f(x) = [x] \left(\frac{\sin \frac{\pi}{[x+3]} + \sin \pi[x+3]}{3+[x]} \right)$ then - **DY0085**

- (A) $f(x)$ is continuous on $\mathbb{R}$ (B) $f(x)$ is continuous but not differentiable on $\mathbb{R}$
(C) $f'(x)$ exists $\forall x \in \mathbb{R}$ (D) $f(x)$ is discontinuous for all integral points in $\mathbb{R}$

EXERCISE (O-2)

Multiple Correct Answer Type

1. If $f(x) = x(\sqrt{x} - \sqrt{x+1})$, then -
 (A) $Rf'(0)$ exist (B) $L f'(0)$ exist but $R f'(0)$ does not exist
 (C) $\lim_{x \rightarrow 0^+} f(x)$ exist (D) $f(x)$ is differentiable at $x = 0$.

DY0020

2. The function $f(x) = \begin{cases} |x-3|, & x \geq 1 \\ \left(\frac{x^2}{4}\right) - \left(\frac{3x}{2}\right) + \left(\frac{13}{4}\right), & x < 1 \end{cases}$ is -
 (A) continuous at $x = 1$ (B) differentiable at $x = 1$
 (C) continuous at $x = 3$ (D) differentiable at $x = 3$

DY0021

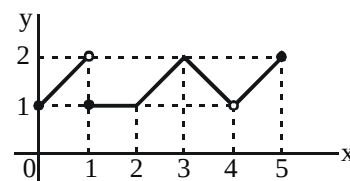
3. Select the correct statements -
 (A) The function f defined by $f(x) = \begin{cases} 2x^2 + 3 & \text{for } x \leq 1 \\ 3x + 2 & \text{for } x > 1 \end{cases}$ is neither differentiable nor continuous at $x = 1$.
 (B) The function $f(x) = x^2|x|$ is twice differentiable at $x = 0$.
 (C) If f is continuous at $x = 5$ and $f(5) = 2$ then $\lim_{x \rightarrow 2} f(4x^2 - 11)$ exists
 (D) If $\lim_{x \rightarrow a} (f(x) + g(x)) = 2$ and $\lim_{x \rightarrow a} (f(x) - g(x)) = 1$ then $\lim_{x \rightarrow a} f(x) \cdot g(x)$ need not exist.

DY0022

4. If $f(x) = \text{sgn}(x^5)$, then which of the following is/are **false** (where sgn denotes signum function) -
 (A) $f'_+(0) = 1$
 (B) $f'_-(0) = -1$
 (C) f is continuous but not differentiable at $x = 0$
 (D) f is discontinuous at $x = 0$

DY0023

5. Graph of $f(x)$ is shown in adjacent figure, then
 (A) $f(x)$ has non removable discontinuity at two points
 (B) $f(x)$ is non differentiable at three points in its domain
 (C) $\lim_{x \rightarrow 1} f(f(x)) = 1$
 (D) Number of points of discontinuity = number of points of non-differentiability



DY0024

6. Let S denotes the set of all points where $\sqrt[5]{x^2|x|^3} - \sqrt[3]{x^2|x|} - 1$ is not differentiable then S is a subset of -
 (A) $\{0, 1\}$ (B) $\{0, 1, -1\}$ (C) $\{0, 1\}$ (D) $\{0\}$

DY0025

7. Which of the following statements is/are correct ?
- (A) There exist a function $f : [0,1] \rightarrow \mathbb{R}$ which is discontinuous at every point in $[0,1]$ & $|f(x)|$ is continuous at every point in $[0,1]$
- (B) Let $F(x) = f(x) \cdot g(x)$. If $f(x)$ is differentiable at $x = a$, $f(a) = 0$ and $g(x)$ is continuous at $x = a$ then $F(x)$ is always differentiable at $x = a$.
- (C) If $Rf'(a) = 2$ & $Lf'(a) = 3$, then $f(x)$ is non-differentiable at $x = a$ but will be always continuous at $x = a$
- (D) If $f(a)$ and $f(b)$ possess opposite signs then there must exist at least one solution of the equation $f(x) = 0$ in (a,b) provided f is continuous on $[a, b]$

DY0026

8. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function. Define $g : \mathbb{R} \rightarrow \mathbb{R}$ by $g(x) = |f(x)|$ for all x . Then which of the following is/are not always true-
- (A) If f is continuous then g is also continuous
- (B) If f is one-one then g is also one-one
- (C) If f is onto then g is also onto
- (D) If f is differentiable then g is also differentiable

DY0027

9. The function $\phi(x) = [x - \sin |x|]$ (where $[.]$ denotes greater integer function) is -
- (A) derivable at $x = 0$
- (B) continuous at $x = 0$
- (C) $\lim_{x \rightarrow 0} \phi(x)$ does not exist
- (D) continuous and derivable at $x = 0$

DY0028

10. Let $f(x) = \begin{cases} x^2 \cos \frac{1}{x}, & x < 0 \\ 0, & x = 0 \\ x^2 \sin \frac{1}{x}, & x > 0 \end{cases}$, then which of the following is (are) correct ?

- (A) $f(x)$ is continuous but not differentiable at $x = 0$
- (B) $f(x)$ is continuous and differentiable at $x = 0$
- (C) $f'(x)$ is continuous but not differentiable at $x = 0$
- (D) $f'(x)$ is discontinuous at $x = 0$

DY0029

11. $f(x) = \begin{cases} a + \sin(\sin x); & x \geq 0 \\ \ln(\cos x) + bx; & x < 0 \end{cases}$ is differentiable at $x = 0$, then -

DY0086

- (A) $a = 1$
- (B) $b = 0$
- (C) $a = 0$
- (D) $b = 1$

12. Let $f(x+y) + f(x-y) = 2f(x) \forall x, y \in \mathbb{R}$, then which of the following is/are be correct ($f(0) = 0$ & f is differentiable $\forall x \in \mathbb{R}$)
- (A) f must be odd
- (B) f must be even
- (C) $f'(1) = f'(2)$
- (D) $f'(2) = f'(3) = f'(4)$

DY0087

13. Let $f(x) = \begin{cases} 1-x, & (0 \leq x \leq 1) \\ x+2, & (1 < x < 2) \\ 4-x, & (2 \leq x \leq 4) \end{cases}$. Then in $0 \leq x \leq 4$ function $y = f(f(x))$ is

DY0088

- (A) continuous at $x = 1$
- (B) not differentiable at $x = 1$
- (C) continuous and differentiable at $x = 2$
- (D) continuous but not differentiable at $x = 3$

14. A function $f(x) = \max(\sin x, \cos x, 1 - \cos x)$ is not derivable for some $x \in [0, 2\pi]$ which may lie in the interval- **DY0089**

(A) $\left[0, \frac{\pi}{2}\right)$ (B) $\left[\pi, \frac{3\pi}{2}\right)$ (C) $\left[\frac{\pi}{2}, \pi\right)$ (D) $\left[\frac{3\pi}{2}, 2\pi\right)$

15. If $f(x) = \begin{cases} ax^2 - b & \text{if } |x| < 1 \\ -\frac{1}{|x|} & \text{if } |x| \geq 1 \end{cases}$ is derivable at $x = 1$, then **DY0090**

(A) $a + b = 2$ (B) $a + 3b = 5$ (C) $a + 2b = 3$ (D) $b - a = 1$

16. Let $g(x) = \begin{cases} a\sqrt{x+2}, & 0 < x < 2 \\ bx + 2, & 2 \leq x < 5 \end{cases}$. If $g(x)$ is derivable on $(0, 5)$, then $(2a + b)$ is greater than or equal to **DY0091**

(A) 1 (B) 2 (C) 3 (D) 4

17. If $f(x) = \begin{cases} c(\tan^{-1}(ax + b)) & x < 0 \\ \frac{1 - \cos(cx)}{x} & x > 0 \end{cases}$, $f(0) = 0$, $(c \neq 0)$ is differentiable at $x = 0$ and $a + b + c < 4$, then a can be **DY0092**

(A) $\frac{1}{6}$ (B) $\frac{1}{2}$ (C) 1 (D) 4

18. Let $f(x) = \min\{\cos x, |x|\}$ is non differentiable at $x = \alpha, \beta$ & γ , where $\alpha < \beta < \gamma$, then which of the following can be true - **DY0093**

(A) $f'(\gamma^+) = -\sqrt{1 - \gamma^2}$ (B) $f'(\beta^-) = 1$
(C) $f'(\alpha^+) = -1$ (D) $f'(\beta^+) = -1$

19. If $\lim_{x \rightarrow 0} \frac{1 - \cos\left(1 - \cos \frac{x}{2}\right)}{2^m x^n}$ is equal to the left hand derivative of $e^{-|x|}$ at $x = 0$, then the value of $|n + 2m|$ is divisible by **DY0094**

(A) 2 (B) 3 (C) 5 (D) 7

20. Consider the function $f(x) = \begin{cases} -2^x & ; x \leq 0 \\ x - k & ; 0 < x < 1 \\ -(x-1)^2 & ; x \geq 1 \end{cases}$. If the function is continuous $\forall x \in \mathbb{R}$, then which of the following holds true? **DY0095**

(A) Range of $f(x)$ is $(-\infty, 0]$ (B) $f(x) = 0$ has exactly one root
(C) function is non-derivable at $x = 0$ & $x = 1$ (D) $f(x)$ is bounded function

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 and 2

A new function tignum (x) (i.e. $\text{tgn}(x)$) is defined such that $\text{tgn}(x) = \begin{cases} 1 & \text{if } [x] = \text{even} \\ -1 & \text{if } [x] = \text{odd} \end{cases}$, where $[.]$

denotes greatest integer function. Let $f(x) = \text{tgn}(x) \sin(x) |x|$.

On the basis of above information, answer the following questions :

- Then number of points in $[0, 10]$ where $f(x)$ is discontinuous is - DY0096
(A) 8 (B) 9 (C) 10 (D) 11
- Then number of points in $[-10, 10]$ where $f(x)$ is non-derivable is - DY0097
(A) 18 (B) 19 (C) 20 (D) 21

Paragraph for Question 3 and 4

Consider the function $f(x) = \begin{cases} x^2 - 5x + 6 & ; -2 \leq x \leq 4 \\ k - \tan\left(\frac{\pi x}{4}\right) & ; 4 < x \leq 5 \\ \log_{10}(\alpha x) & ; 5 < x \leq 10 \end{cases}$. Let $f_1(x) = |f(x)|$.

On the basis of above information, answer the following questions :

- If $f(x)$ is continuous in $[-2, 10]$ then - DY0098
(A) $\alpha = 2$ (B) $\alpha = 4$ (C) $k = 2$ (D) $k = 4$
- If $f(x)$ is continuous in $[-2, 10]$, then the number of points, where $f_1(x)$ is NOT differentiable, is - DY0099
(A) 9 (B) 8 (C) 7 (D) 5

Paragraph for Question 5 and 6

Consider $f(x) = [3x + 4] + |4x^2 - 1|(2x^2 + 3x - 2) + \sin\left(\frac{\pi x}{2}\right)$ in $[-4, 4]$

(where $[.]$ denotes greatest integer function) is non-derivable

- Number of points where $f(x)$ is discontinuous is (are) DY0100
- Number of points where $f(x)$ is non derivable is (are) DY0101

Paragraph for Question 7 and 8

Let $f(x) = \begin{cases} \min\{\sin x, \cos x\} & ; -\frac{\pi}{2} \leq x < 0 \\ \max\{\sin x, \cos x\} & ; 0 \leq x \leq \frac{\pi}{2} \end{cases}$, then

- Number of points where $f(x)$ is discontinuous and non differentiable is M & N respectively in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, then $(M + N)$ is - DY0102
- Number of solutions of equation $2f(x) = 1$ is - DY0103

Matrix Match Type

- | 9. | List-I | List-II |
|----|--|---------|
| | (P) If $f(x)$ is derivable at $x = 3$ & $f'(3) = 2$,
then $\lim_{h \rightarrow 0} \frac{f(3+h^2) - f(3-h^2)}{2h^2}$ equals | (1) 0 |
| | (Q) Let $f(x)$ be a function satisfying the condition
$f(-x) = f(x)$ for all real x . If $f'(0)$ exists, then its
value is equal to | (2) 1 |
| | (R) For the function $f(x) = \begin{cases} \frac{x}{1+e^{1/x}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$
the derivative from the left $Lf'(0)$ equals | (3) 2 |
| | (S) The number of points at which the function
$f(x) = \max. \{a - x, a + x, b\}$, $-\infty < x < \infty$,
$0 < a < b$ cannot be differentiable is | (4) 3 |

DY0030

- (A) (P) $\rightarrow$ 3, (Q) $\rightarrow$ 2, (R) $\rightarrow$ 1, (S) $\rightarrow$ 3 (B) (P) $\rightarrow$ 4, (Q) $\rightarrow$ 1, (R) $\rightarrow$ 2, (S) $\rightarrow$ 4
 (C) (P) $\rightarrow$ 3, (Q) $\rightarrow$ 1, (R) $\rightarrow$ 2, (S) $\rightarrow$ 3 (D) (P) $\rightarrow$ 2, (Q) $\rightarrow$ 2, (R) $\rightarrow$ 1, (S) $\rightarrow$ 3

10. Given $f(x) = \text{sgn}(x^2 - 4|x| + \lambda)$ (where $\text{sgn}(x)$ denotes signum function of x , $\lambda \in \mathbb{R}$) **DY0104**
 In the column-I there are values of λ and in the column-II there are number of points where $f(x)$ is non-differentiable.

- | Column-I | Column-II |
|-------------------|-----------|
| (A) $\lambda = 0$ | (P) 0 |
| (B) $\lambda = 4$ | (Q) 3 |
| (C) $\lambda = 5$ | (R) 4 |
| (D) $\lambda = 2$ | (S) 2 |

EXERCISE (O-4)

Numerical Grid Type

1. If the function $f(x)$ defined as $f(x) = \begin{cases} -\frac{x^2}{2} & \text{for } x \leq 0 \\ x^n \sin \frac{1}{x} & \text{for } x > 0 \end{cases}$ is continuous but not derivable at $x = 0$ then sum of all non-negative integral value(s) of 'n' is **DY0105**
2. If $f(x) = |x - 1| \cdot ([x] - [-x])$, where $[x]$ denotes greatest integer function, then the value of $(Rf'(1) + Lf'(1))$ is **DY0106**
3. Let $f(x)$ be defined in the interval $[-2, 2]$ such that $f(x) = \begin{cases} -1, & -2 \leq x \leq 0 \\ x-1, & 0 < x \leq 2 \end{cases}$ & $g(x) = f(|x|) + |f(x)|$, then the number of points of non-differentiability of $g(x)$ in $(-2, 2)$ is **DY0107**
4. Let $f(x) = \begin{cases} |2x - 3|[x] & \text{for } x \geq 1 \\ \sin \frac{\pi x}{2} & \text{for } x < 1 \end{cases}$ **DY0108**
where $[.]$ denotes the greatest integer function, then the number of points of non-differentiability of $f(x)$ in $[0, 2]$ is
5. For any real number x , let $[x]$ denote the largest integer less than or equal to x . Let f be a real valued function defined on the interval $[-3, 3]$ by $f(x) = \begin{cases} -x - [-x] & \text{if } [x] \text{ is even} \\ x - [x] & \text{if } [x] \text{ is odd} \end{cases}$
If L denotes the number of point of discontinuity and M denotes the number of points of non-derivability of $f(x)$, then find $(L + M)$. **DY0044**
6. A derivable function $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ satisfies the condition $f(x) - f(y) \geq \ln(x/y) + x - y$ for every $x, y \in \mathbb{R}^+$. If g denotes the derivative of f then compute the value of the sum $\sum_{n=1}^{100} g\left(\frac{1}{n}\right)$. **DY0046**
7. Let $f(x) = [[x] + \{x^2\}] + \{[x^2] + \{x\}\}$, then number of points where $|f(x)|$ is non-derivable in $[-3, 3]$ is equal to (where $[.]$ denotes greatest integer function & $\{.\}$ denotes fractional part function) **DY0109**
8. If f is a differentiable function such that $f\left(\frac{x+y}{3}\right) = \frac{f(x) + f(y) + f(0)}{3}$, $\forall x, y \in \mathbb{R}$ and $f'(0) = 2$, then the slope of tangent to $f(x)$ at $x = 7$ is **DY0110**
9. Let $f(x)$ be a differentiable function such that $\lim_{x \rightarrow 0} \frac{f(3 - \sin x) - f(3 + x)}{x} = 8$, then $|f'(3)|$ is **DY0049**
10. Let $f(x)$ be a differentiable function such that $2f(x + y) + f(x - y) = 3f(x) + 3f(y) + 2xy$ $\forall x, y \in \mathbb{R}$ & $f'(0) = 0$, then $f(10) + f'(10)$ is equal to **DY0050**

EXERCISE (JM)

1. Let $f(x) = x|x|$ and $g(x) = \sin x$. [AIEEE-2009]

Statement-1 : $g \circ f$ is differentiable at $x = 0$ and its derivative is continuous at that point.

Statement-2 : $g \circ f$ is twice differentiable at $x = 0$.

- (1) Statement-1 is true, Statement-2 is false.
 (2) Statement-1 is false, Statement-2 is true.
 (3) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.
 (4) Statement-1 is true, Statement-2 is true ; Statement-2 is not a correct explanation for statement-1.

DY0051

2. If function $f(x)$ is differentiable at $x = a$ then $\lim_{x \rightarrow a} \frac{x^2 f(a) - a^2 f(x)}{x - a}$ [AIEEE-2011]

- (1) $2a f(a) + a^2 f'(a)$ (2) $-a^2 f'(a)$ (3) $af(a) - a^2 f'(a)$ (4) $2af(a) - a^2 f'(a)$

DY0052

3. Consider the function,
 $f(x) = |x - 2| + |x - 5|$, $x \in \mathbb{R}$.

Statement-1 : $f'(4) = 0$.

Statement-2 : f is continuous in $[2, 5]$, differentiable in $(2, 5)$ and $f(2) = f(5)$. [AIEEE 2012]

- (1) Statement-1 is true, Statement-2 is false.
 (2) Statement-1 is false, Statement-2 is true.
 (3) Statement-1 is true, Statement-2 is true ; Statement-2 is a correct explanation for Statement1.
 (4) Statement-1 is true, Statement-2 is true ; Statement-2 is not a correct explanation for Statement1.

DY0053

4. Let $f(x) = x|x|$, $g(x) = \sin x$ and $h(x) = (g \circ f)(x)$. Then [On-line 2014]

- (1) $h'(x)$ is differentiable at $x = 0$
 (2) $h'(x)$ is continuous at $x = 0$ but is not differentiable at $x = 0$
 (3) $h(x)$ is differentiable at $x = 0$ but $h'(x)$ is not continuous at $x = 0$
 (4) $h(x)$ is not differentiable at $x = 0$

DY0054

5. Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be two functions defined by $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$, and $g(x) = xf(x) :-$

Statement I : f is a continuous function at $x = 0$.

[On-line 2014]

Statement II : g is a differentiable function at $x = 0$.

- (1) Statement I is false and statement II is true
 (2) Statement I is true and statement II is false
 (3) Both statement I and II are true
 (4) Both statements I and II are false

DY0055

6. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $|f(x)| \leq x^2$, for all $x \in \mathbb{R}$. Then, at $x = 0$, f is :
 (1) Neither continuous nor differentiable
 (2) differentiable but not continuous
 (3) continuous as well as differentiable
 (4) continuous but not differentiable

[On-line 2014]

DY0056

7. For $x \in \mathbb{R}$, $f(x) = |\log 2 - \sin x|$ and $g(x) = f(f(x))$, then :

[JEE(Main)-2016]

- (1) g is differentiable at $x = 0$ and $g'(0) = -\sin(\log 2)$
 (2) g is not differentiable at $x = 0$
 (3) $g'(0) = \cos(\log 2)$
 (4) $g'(0) = -\cos(\log 2)$

DY0057

8. Let $S = \{t \in \mathbb{R} : f(x) = |x - \pi| \cdot (e^{|x|} - 1) \sin|x| \text{ is not differentiable at } t\}$. Then the set S is equal to :

[JEE(Main)-2018]

- (1) $\{0\}$ (2) $\{\pi\}$ (3) $\{0, \pi\}$ (4) ϕ (an empty set)

DY0058

9. Let f be a differentiable function from $\mathbb{R}$ to $\mathbb{R}$ such that $|f(x) - f(y)| \leq 2|x - y|^{\frac{3}{2}}$, for all $x, y \in \mathbb{R}$.

If $f(0) = 1$ then $\int_0^1 f^2(x) dx$ is equal to

[JEE(Main)-2019]

- (1) 0 (2) $\frac{1}{2}$ (3) 2 (4) 1

DY0059

10. Let $f(x) = \begin{cases} \max\{|x|, x^2\}, & |x| \leq 2 \\ 8 - 2|x|, & 2 < |x| \leq 4 \end{cases}$. Let S be the set of points in the interval $(-4, 4)$ at which f is not differentiable. Then S :

[JEE(Main)-2019]

- (1) is an empty set (2) equals $\{-2, -1, 1, 2\}$
 (3) equals $\{-2, -1, 0, 1, 2\}$ (4) equals $\{-2, 2\}$

DY0060

11. Let $f : (-1, 1) \rightarrow \mathbb{R}$ be a function defined by $f(x) = \max\{-|x|, -\sqrt{1 - x^2}\}$. If K be the set of all points at which f is not differentiable, then K has exactly :

[JEE(Main)-2019]

- (1) Three elements (2) One element (3) Five elements (4) Two elements

DY0061

12. Let K be the set of all real values of x where the function $f(x) = \sin|x| - |x| + 2(x - \pi) \cos|x|$ is not differentiable. Then the set K is equal to :-

[JEE(Main)-2019]

- (1) $\{\pi\}$ (2) $\{0\}$ (3) ϕ (an empty set) (4) $\{0, \pi\}$

DY0062

13. Let S be the set of all points in $(-\pi, \pi)$ at which the function, $f(x) = \min \{\sin x, \cos x\}$ is not differentiable. Then S is a subset of which of the following? [JEE(Main)-2019]

(1) $\left\{-\frac{3\pi}{4}, -\frac{\pi}{4}, \frac{3\pi}{4}, \frac{\pi}{4}\right\}$ (2) $\left\{-\frac{3\pi}{4}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{4}\right\}$ (3) $\left\{-\frac{\pi}{2}, -\frac{\pi}{4}, \frac{\pi}{4}, \frac{\pi}{2}\right\}$ (4) $\left\{-\frac{\pi}{4}, 0, \frac{\pi}{4}\right\}$

DY0063

14. Let $f(x) = 15 - |x - 10|$; $x \in \mathbb{R}$. Then the set of all values of x , at which the function, $g(x) = f(f(x))$ is not differentiable, is : [JEE(Main)-2019]

(1) $\{5, 10, 15, 20\}$ (2) $\{10, 15\}$ (3) $\{5, 10, 15\}$ (4) $\{10\}$

DY0064

15. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable at $c \in \mathbb{R}$ and $f(c) = 0$. If $g(x) = |f(x)|$, then at $x = c$, g is : [JEE(Main)-2019]

(1) differentiable if $f'(c) = 0$ (2) not differentiable
(3) differentiable if $f'(c) \neq 0$ (4) not differentiable if $f'(c) = 0$

DY0065

16. Suppose a differentiable function $f(x)$ satisfies the identity $f(x+y) = f(x) + f(y) + xy^2 + x^2y$, for all real x and y . If $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$, then $f'(3)$ is equal to _____. [JEE(Main)-2020]

DY0111

17. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as $f(x) = \begin{cases} 3\left(1 - \frac{|x|}{2}\right) & \text{if } |x| \leq 2 \\ 0 & \text{if } |x| > 2 \end{cases}$

Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be given by $g(x) = f(x+2) - f(x-2)$. If n and m denote the number of points in $\mathbb{R}$ where g is not continuous and not differentiable, respectively, then $n + m$ is equal to _____.

[JEE(Main)-2021]

DY0112

18. The number of points, where the function $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = |x-1| \cos |x-2| \sin |x-1| + (x-3)|x^2 - 5x + 4|$, is NOT differentiable, is : [JEE(Main)-2022]

(A) 1 (B) 2 (C) 3 (D) 4

DY0113

EXERCISE (JA)

1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that

$$f(x+y) = f(x) + f(y), \forall x, y \in \mathbb{R}.$$

If $f(x)$ is differentiable at $x = 0$, then

- (A) $f(x)$ is differentiable only in a finite interval containing zero
 (B) $f(x)$ is continuous $\forall x \in \mathbb{R}$
 (C) $f'(x)$ is constant $\forall x \in \mathbb{R}$
 (D) $f(x)$ is differentiable except at finitely many points

[JEE 2011, 4M]

DY0066

2. If $f(x) = \begin{cases} -x - \frac{\pi}{2} & , \quad x \leq -\frac{\pi}{2} \\ -\cos x & , \quad -\frac{\pi}{2} < x \leq 0 \\ x-1 & , \quad 0 < x \leq 1 \\ \ln x & , \quad x > 1 \end{cases}$ then -

[JEE 2011, 4M]

- (A) $f(x)$ is continuous at $x = -\frac{\pi}{2}$
 (B) $f(x)$ is not differentiable at $x = 0$
 (C) $f(x)$ is differentiable at $x = 1$
 (D) $f(x)$ is differentiable at $x = -\frac{3}{2}$

DY0067

3. Let $f(x) = \begin{cases} x^2 \left| \cos \frac{\pi}{x} \right| & , \quad x \neq 0 \\ 0 & , \quad x = 0 \end{cases}$, $x \in \mathbb{R}$, then f is -

[JEE 2012, 3M, -1M]

- (A) differentiable both at $x = 0$ and at $x = 2$
 (B) differentiable at $x = 0$ but not differentiable at $x = 2$
 (C) not differentiable at $x = 0$ but differentiable at $x = 2$
 (D) differentiable neither at $x = 0$ nor at $x = 2$

DY0068

4. Let $f_1 : \mathbb{R} \rightarrow \mathbb{R}$, $f_2 : [0, \infty) \rightarrow \mathbb{R}$, $f_3 : \mathbb{R} \rightarrow \mathbb{R}$ and $f_4 : \mathbb{R} \rightarrow [0, \infty)$ be defined by

$$f_1(x) = \begin{cases} |x| & \text{if } x < 0, \\ e^x & \text{if } x \geq 0; \end{cases}$$

$$f_2(x) = x^2;$$

$$f_3(x) = \begin{cases} \sin x & \text{if } x < 0, \\ x & \text{if } x \geq 0 \end{cases}$$

$$\text{and } f_4(x) = \begin{cases} f_2(f_1(x)) & \text{if } x < 0, \\ f_2(f_1(x)) - 1 & \text{if } x \geq 0. \end{cases}$$

List-I

- P. f_4 is
Q. f_3 is
R. $f_2 \circ f_1$ is
S. f_2 is

List-II

1. onto but not one-one
2. neither continuous nor one-one
3. differentiable but not one-one
4. continuous and one-one

Codes :

	P	Q	R	S
(A)	3	1	4	2
(B)	1	3	4	2
(C)	3	1	2	4
(D)	1	3	2	4

[JEE(Advanced)-2014, 3(-1)]

DY0069

5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be respectively given by $f(x) = |x| + 1$ and $g(x) = x^2 + 1$. Define

$$h : \mathbb{R} \rightarrow \mathbb{R} \text{ by } h(x) = \begin{cases} \max\{f(x), g(x)\} & \text{if } x \leq 0, \\ \min\{f(x), g(x)\} & \text{if } x > 0. \end{cases}$$

The number of points at which $h(x)$ is not differentiable is

[JEE(Advanced)-2014, 3]

DY0070

6. Let $a, b \in \mathbb{R}$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = a \cos(|x^3 - x|) + b|x| \sin(|x^3 + x|)$. Then f is -

- (A) differentiable at $x = 0$ if $a = 0$ and $b = 1$
(B) differentiable at $x = 1$ if $a = 1$ and $b = 0$
(C) **NOT** differentiable at $x = 0$ if $a = 1$ and $b = 0$
(D) **NOT** differentiable at $x = 1$ if $a = 1$ and $b = 1$

[JEE(Advanced)-2016, 4(-2)]

DY0071

7. Let $f : \left[-\frac{1}{2}, 2\right] \rightarrow \mathbb{R}$ and $g : \left[-\frac{1}{2}, 2\right] \rightarrow \mathbb{R}$ be function defined by $f(x) = [x^2 - 3]$ and

$g(x) = |x| f(x) + |4x - 7| f(x)$, where $[y]$ denotes the greatest integer less than or equal to y for $y \in \mathbb{R}$. Then

(A) f is discontinuous exactly at three points in $\left[-\frac{1}{2}, 2\right]$

(B) f is discontinuous exactly at four points in $\left[-\frac{1}{2}, 2\right]$

(C) g is NOT differentiable exactly at four points in $\left(-\frac{1}{2}, 2\right)$

(D) g is NOT differentiable exactly at five points in $\left(-\frac{1}{2}, 2\right)$

[JEE(Advanced)-2016, 4(-2)]

DY0072

8. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = 1$ and satisfying the equation

$$f(x+y) = f(x)f'(y) + f'(x)f(y) \text{ for all } x, y \in \mathbb{R}.$$

Then, then value of $\log_e(f(4))$ is _____.

[JEE(Advanced)-2018]

DY0073

9. Let $f_1 : \mathbb{R} \rightarrow \mathbb{R}$, $f_2 : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$, $f_3 : \left(-1, e^{\frac{\pi}{2}} - 2\right) \rightarrow \mathbb{R}$ and $f_4 : \mathbb{R} \rightarrow \mathbb{R}$ be functions defined by

$$(i) f_1(x) = \sin\left(\sqrt{1-e^{-x^2}}\right)$$

$$(ii) f_2(x) = \begin{cases} \frac{|\sin x|}{\tan^{-1} x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}, \text{ where the inverse trigonometric function } \tan^{-1} x \text{ assumes values in } \left(-\frac{\pi}{2}, \frac{\pi}{2}\right),$$

$$(iii) f_3(x) = [\sin(\log_e(x+2))], \text{ where for } t \in \mathbb{R}, [t] \text{ denotes the greatest integer less than or equal to } t,$$

$$(iv) f_4(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

List-I

- P. The function f_1 is
Q. The function f_2 is
R. The function f_3 is
S. The function f_4 is

List-II

1. **NOT** continuous at $x = 0$
2. continuous at $x = 0$ and **NOT** differentiable at $x = 0$
3. differentiable at $x = 0$ and its derivative is **NOT** continuous at $x = 0$
4. differentiable at $x = 0$ and its derivative is continuous at $x = 0$

The correct option is :

- (A) P $\rightarrow$ 2; Q $\rightarrow$ 3; R $\rightarrow$ 1; S $\rightarrow$ 4
(B) P $\rightarrow$ 4; Q $\rightarrow$ 1; R $\rightarrow$ 2; S $\rightarrow$ 3
(C) P $\rightarrow$ 4; Q $\rightarrow$ 2; R $\rightarrow$ 1; S $\rightarrow$ 3
(D) P $\rightarrow$ 2; Q $\rightarrow$ 1; R $\rightarrow$ 4; S $\rightarrow$ 3

[JEE(Advanced)-2018]

DY0074

10. Let the function $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^3 - x^2 + (x - 1) \sin x$ and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be an arbitrary function. Let $fg : \mathbb{R} \rightarrow \mathbb{R}$ be the product function defined by $(fg)(x) = f(x)g(x)$. Then which of the following statements is/are TRUE ? [JEE(Advanced)-2020]

- (A) If g is continuous at $x = 1$, then fg is differentiable at $x = 1$
 (B) If fg is differentiable at $x = 1$, then g is continuous at $x = 1$
 (C) If g is differentiable at $x = 1$, then fg is differentiable at $x = 1$
 (D) If fg is differentiable at $x = 1$, then g is differentiable at $x = 1$

DY0075

11. Let the functions $f : (-1, 1) \rightarrow \mathbb{R}$ and $g : (-1, 1) \rightarrow (-1, 1)$ be defined by

$$f(x) = |2x - 1| + |2x + 1| \text{ and } g(x) = x - [x],$$

where $[x]$ denotes the greatest integer less than or equal to x . Let $f \circ g : (-1, 1) \rightarrow \mathbb{R}$ be the composite function defined by $(f \circ g)(x) = f(g(x))$. Suppose c is the number of points in the interval $(-1, 1)$ at which $f \circ g$ is **NOT** continuous, and suppose d is the number of points in the interval $(-1, 1)$ at which $f \circ g$ is **NOT** differentiable. Then the value of $c + d$ is _____. [JEE(Advanced)-2020]

DY0076

12. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be functions satisfying $f(x + y) = f(x) + f(y) + f(x)f(y)$ and $f(x) = xg(x)$ for all $x, y \in \mathbb{R}$. If $\lim_{x \rightarrow 0} g(x) = 1$, then which of the following statements is/are TRUE ? [JEE(Advanced)-2020]

- (A) f is differentiable at every $x \in \mathbb{R}$
 (B) If $g(0) = 1$, then g is differentiable at every $x \in \mathbb{R}$
 (C) The derivative $f'(1)$ is equal to 1
 (D) The derivative $f'(0)$ is equal to 1

DY0077

ANSWER KEY

Do yourself - 1

- Continuous but not differentiable at $x = 1$
- $a = \frac{1}{6}, b = \frac{\pi}{4} - \frac{13}{6}$
- (a) DNE (b) DNE (c) DNE
- $a = \frac{1}{e^2}, b = 1$
- $b = -1, c = 1$
 $= 1$
- $f(x)$ is continuous at $x = 1$ but not differentiable at x

Do yourself - 2

- Continuous & differentiable at $x = 1$
- continuous on $\mathbb{R} - \{1\}$ and differentiable on $\mathbb{R} - \{-1, 1\}$.
- (D)

Do yourself - 3

- Continuous everywhere but not differentiable at $x = 2$ only
- 5
- (7, 3)
- $x = \pi - \cos^{-1} \frac{3}{5}$
- Not differentiable at $x = \pm 1$

Do yourself - 4

- 3
- (a) 1 & 2 (b) Not continuous at $x = 1$ & 2 and not differentiable at $x = 1$ & 2.
- $f(x)$ is not differentiable at $x = \frac{1}{2}, 1$
- Non derivable at $x = 0, 1, 2$
- Non derivable at $x = 0$, continuous at $x = 0$
derivable and continuous at $x = \frac{\pi}{2}$

Do yourself - 5

- $f(x) = x + \frac{x^3}{3}$
- $f(x) = e^{2x}$
- $f(x) = mx$

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	A	C	B	D	A	B	D	B	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	D	D	A	D	D	D	B	A	B
Que.	21	22	23	24	25					
Ans.	D	B	C	C	D					

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C,D	A,B,C	B,C	A,B,C	B,C	A,B,C,D	A,B,C,D	B,C,D	A,B,D	B,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C,D	A,C,D	A,B	A,C,D	A,B,D	A,B,C	A,B,C	A,C	A,C	A,B,C

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9	
Ans.	C	B	A,C	A	24.00	25.00	3.00	0.00	C	
Q.10	A	B	C	D						
	Q	S	P	R						

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	2	2	3	8	5150.00	1	2.00	4.00	120.00

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	4	4	2	3	3	3	4	4	3
Que.	11	12	13	14	15	16	17	18		
Ans.	1	3	1	3	1	10	4	B		

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B,C	A,B,C,D	B	D	3	A,B	B,C	2	D	A,C
Que.	11	12								
Ans.	4.00	A,B,D								